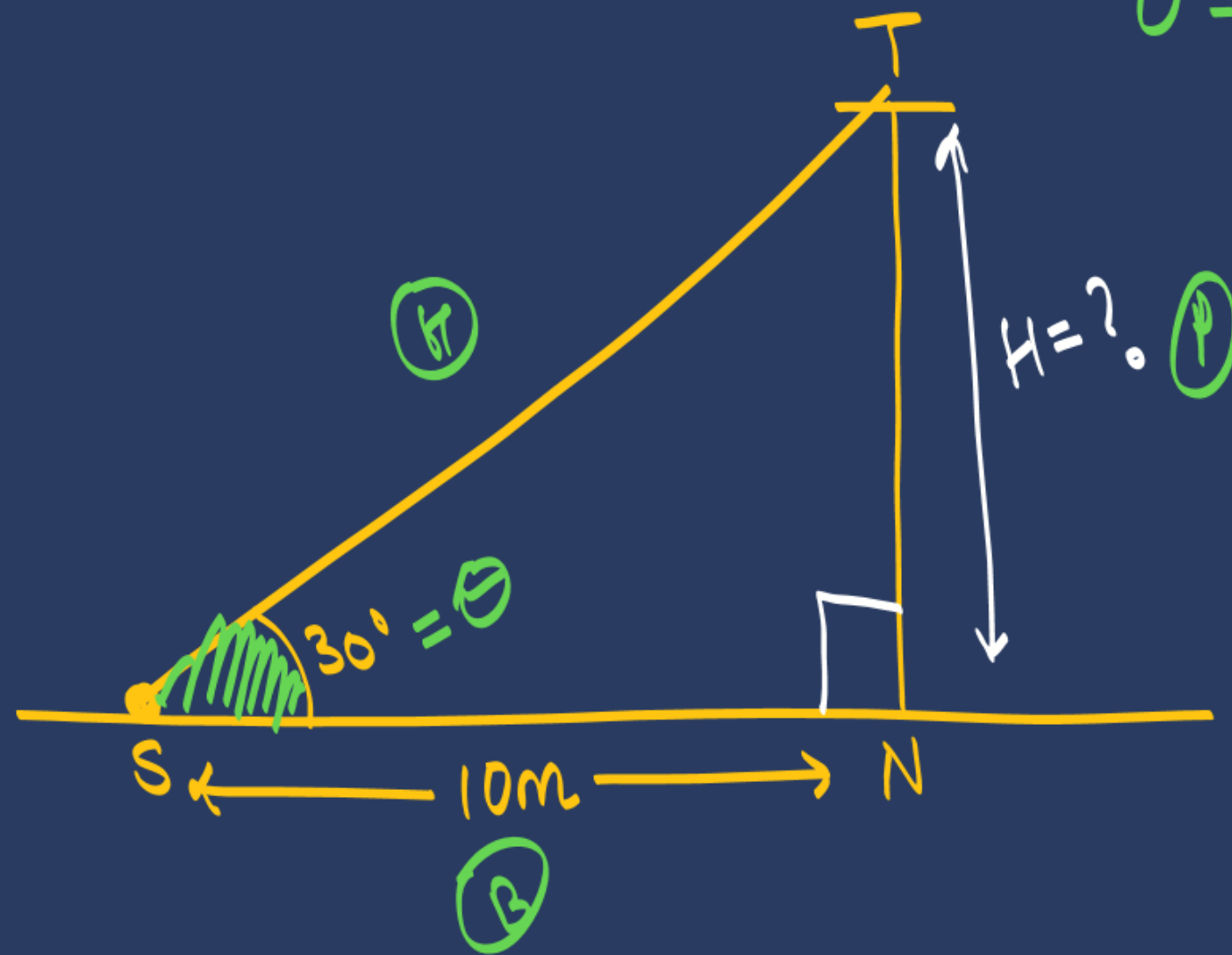


#14Days14Chapters

DAY 3

[Aim: 100/100 in Maths]

SOME APPLICATIONS OF TRIGONOMETRY



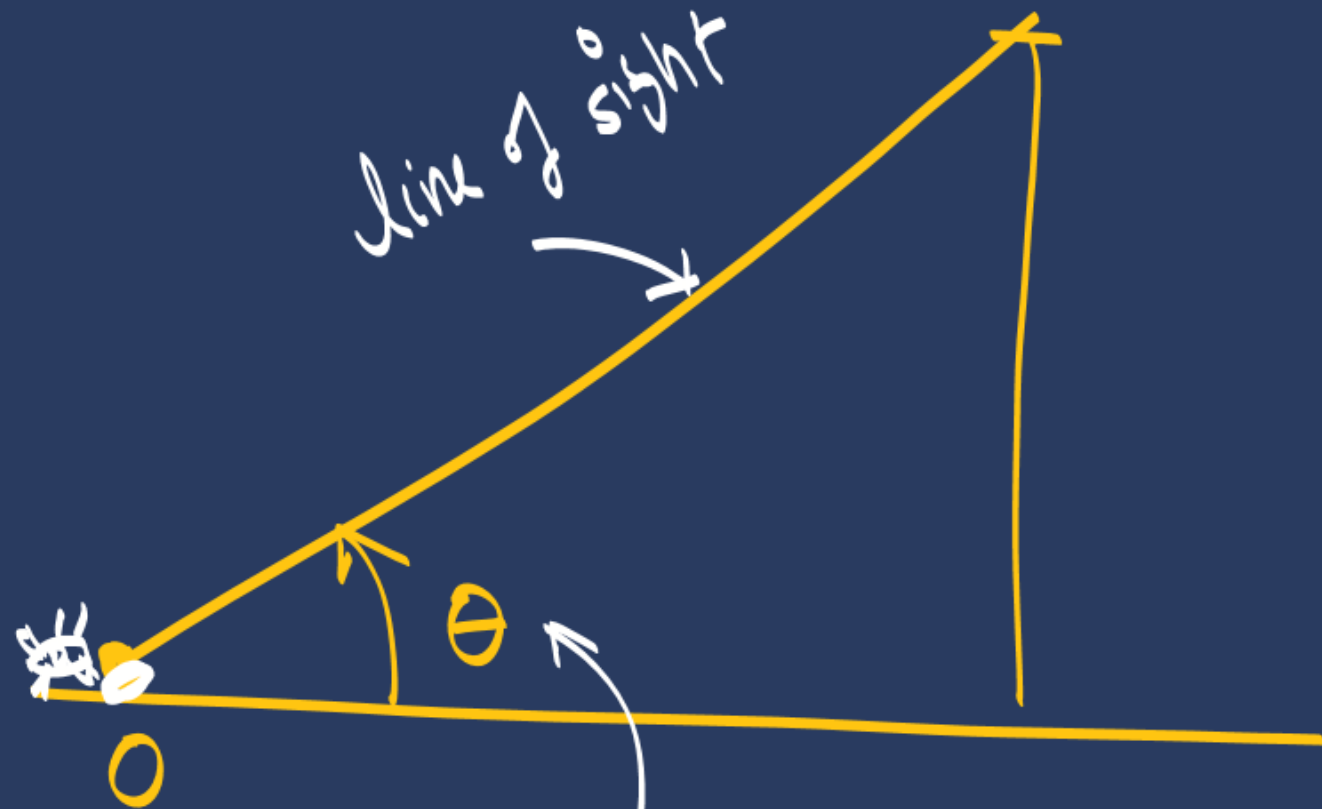
$$\theta = 30^\circ, \text{ B} \checkmark, \text{ ? ?}$$

$$\tan 30 = \frac{NT}{SN}$$

$$\frac{1}{\sqrt{3}} = \frac{H}{10}$$

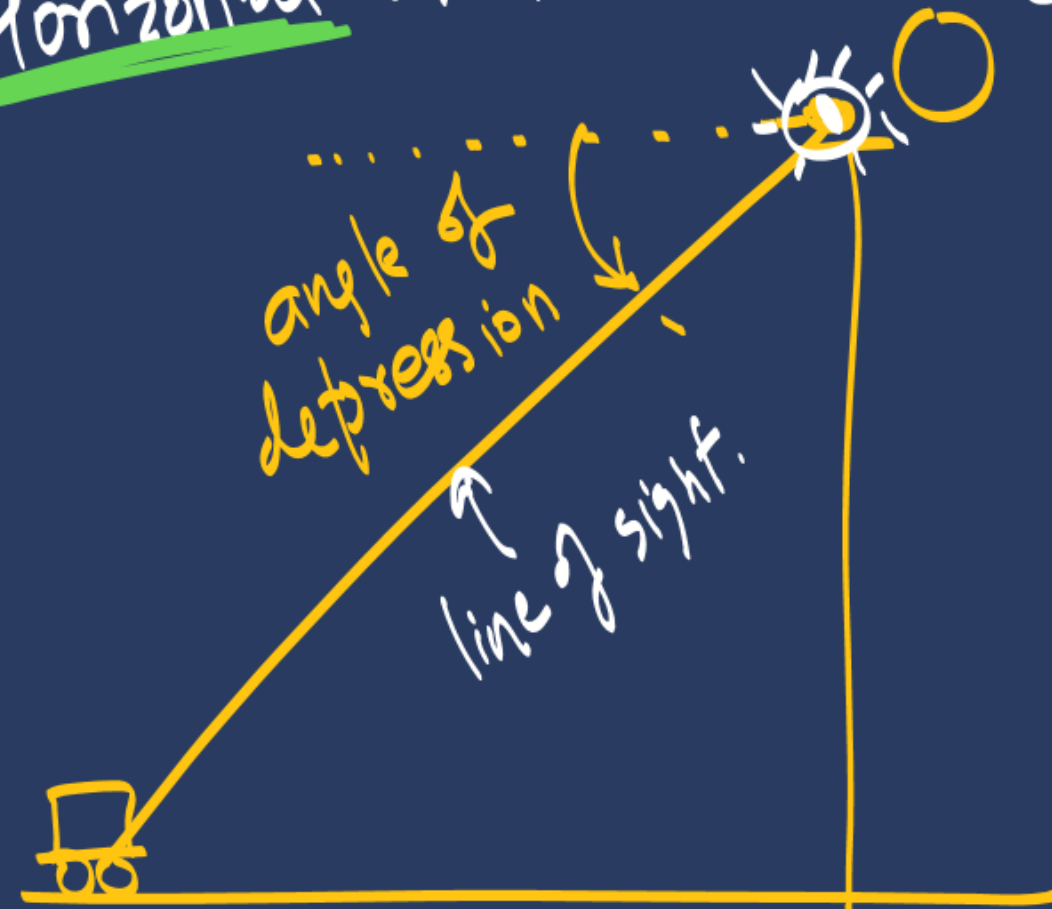
$$H = \frac{10}{\sqrt{3}} \text{ m} \checkmark \checkmark$$

#(K3B)



Horizontal से कितना ऊपर रहे।

Horizontal से कितना नीचे।



⊛ Assume observer as point object unless given.

#LP: From a point on the ground, which is 30 m away from the foot of a vertical tower, the angle of elevation of the top of the tower is found to be 60° . The height (in metres) of the tower is:

(CBSE 2024)

(a) $10\sqrt{3}$

☒ (b) $30\sqrt{3}$

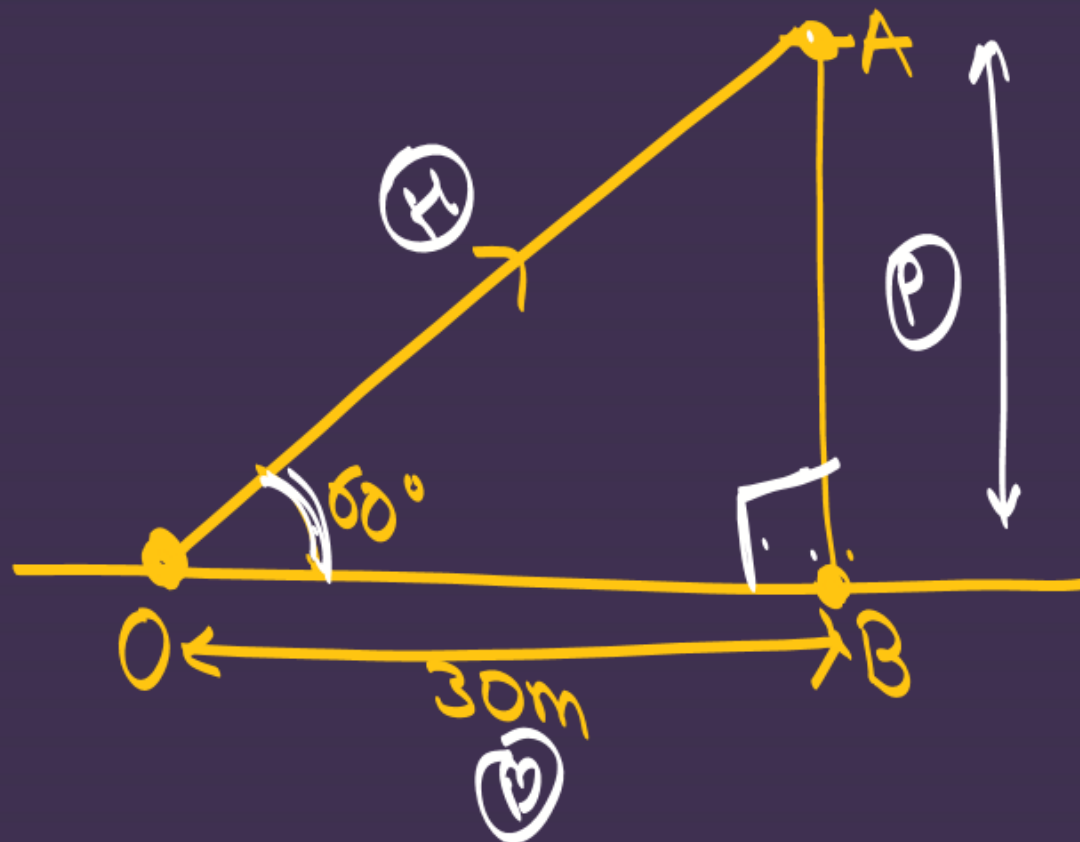
(c) 60

(d) 30

Let AB is the tower, O is the point which is 30m away from foot of tower.

Since tower stands vertically on ground. $\therefore \angle ABO = 90^\circ$

We have to find AB.



Now In right $\triangle ABO$,

$$\tan 60 = \frac{AB}{OB}$$

$$\sqrt{3} = \frac{AB}{30}$$

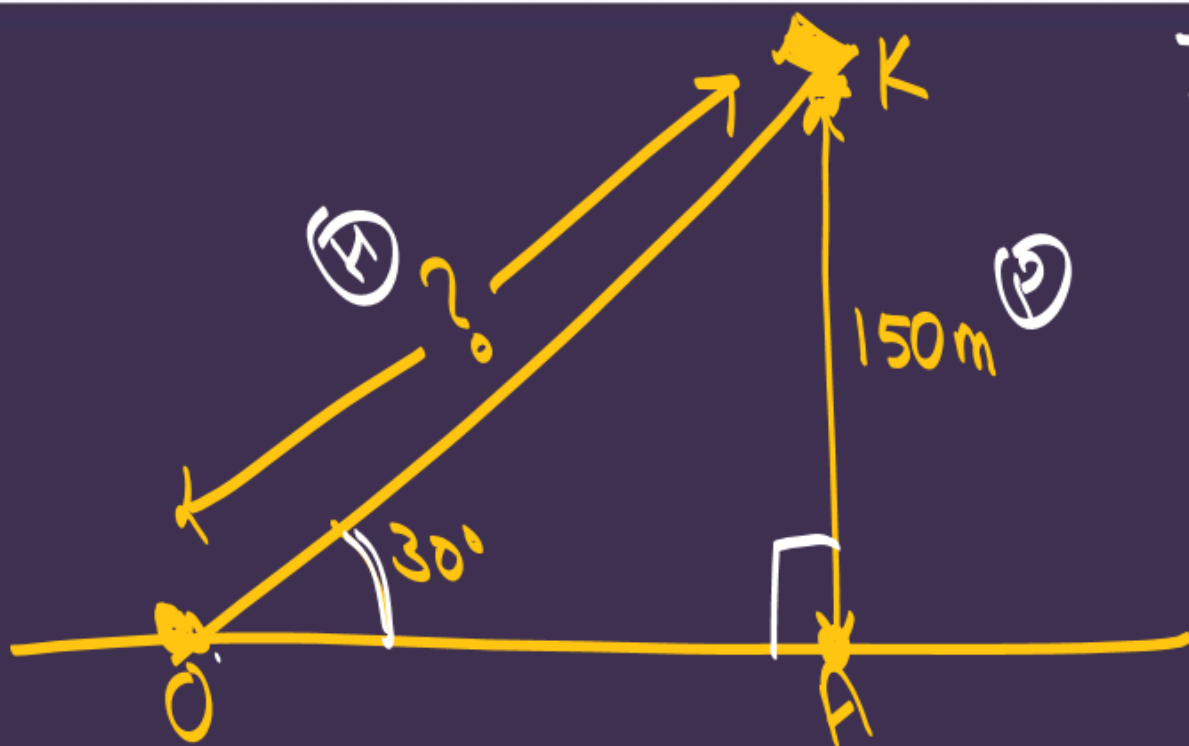
$$AB = 30\sqrt{3}m$$

$\therefore$ height of the tower is $30\sqrt{3}m$.

#LP: A kite is flying at a height of 150 m from the ground. It is attached to a string inclined at an angle of 30° to the horizontal. The length of the string is : **(1 Mark)**
(2025)

$$\frac{P}{H} = \sin \theta$$

- (A) 100 m
- ✓ (B) 300 m
- (C) 150 m
- (D) 150 m



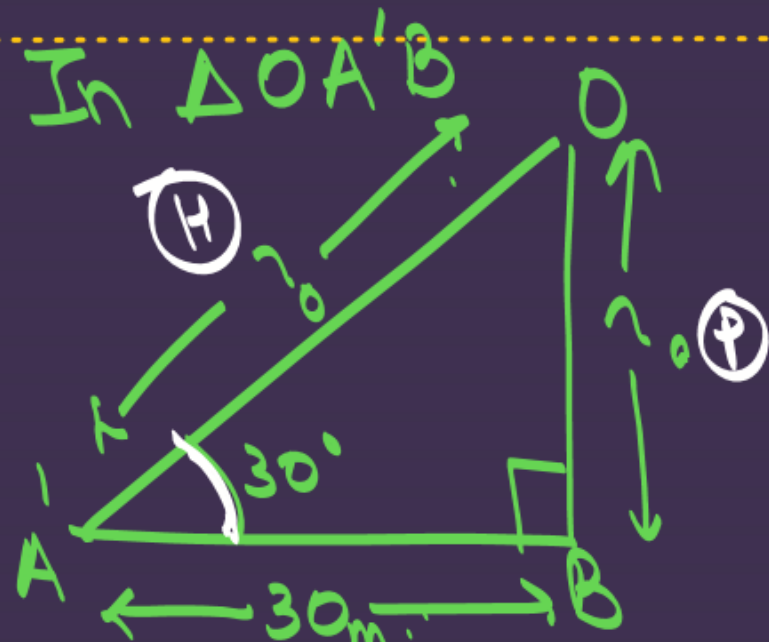
In $\triangle OKA$, $\sin 30^\circ = \frac{150}{OK}$

$$\frac{1}{2} = \frac{150}{OK} \Rightarrow \boxed{OK = 300 \text{ m}}$$

#LP: A tree is broken by wind. The top struck the ground at an angle of 30° and at a distance of 30m from the root. Find the height of the whole tree.

To find $[AB] = OA + OB$

$$AB = OA' + OB \Rightarrow (20\sqrt{3}m) + (10\sqrt{3}m) \Rightarrow \underline{\underline{30\sqrt{3}m}}$$



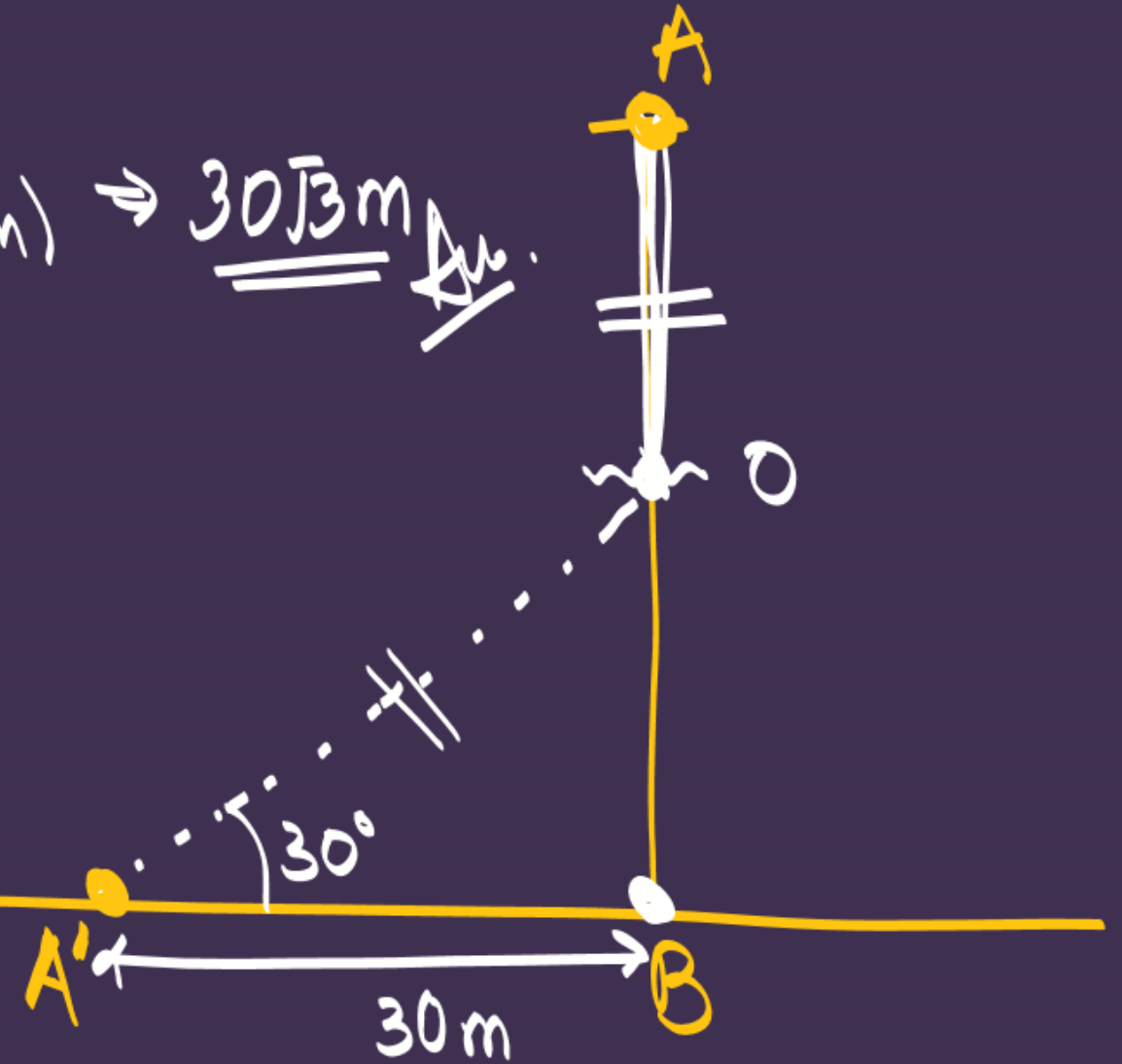
$$\cos 30 = \frac{AB}{OA'}$$

$$\frac{\sqrt{3}}{2} = \frac{30}{OA'}$$

$$OA' = \frac{30 \times 2}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{60\sqrt{3}}{3} = 20\sqrt{3}m = OA'$$

$$\tan 30 = \frac{OB}{30}$$

$$\frac{1}{\sqrt{3}} = \frac{OB}{30} \Rightarrow OB = \frac{30}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{30\sqrt{3}}{3} = 10\sqrt{3} = OB$$



#LP: An observer 1.5 m tall is 28.5 m away from a tower. The angle of elevation of the top of the tower from his/her eyes has measure 45° . What is the height of the tower?

A. 28.5 m

☒ B. 30 m

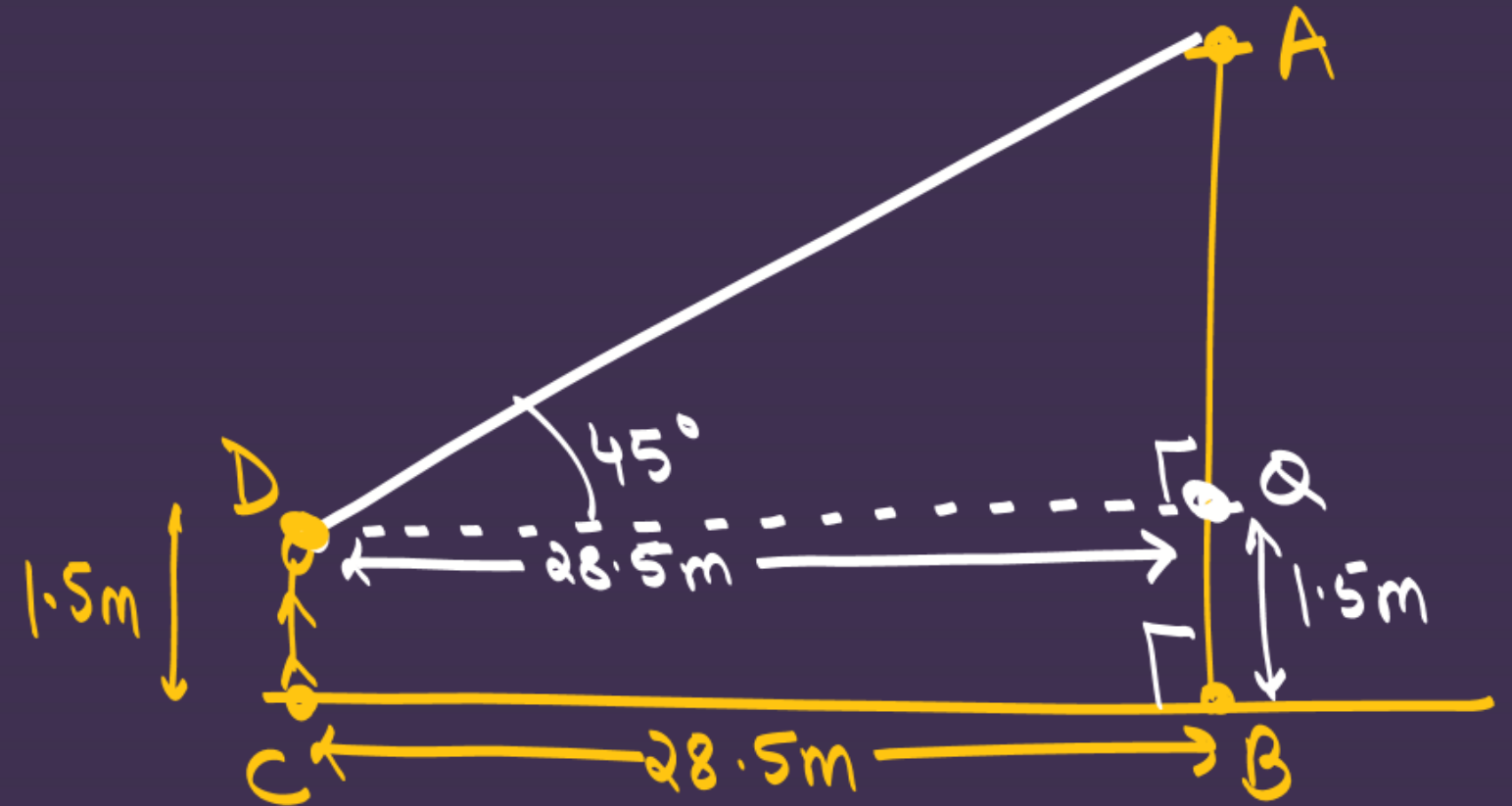
C. 27 m

D. 1.5 m

In $\triangle A Q D$, $\tan 45 = \frac{A Q}{28.5}$

$$1 = \frac{A Q}{28.5}$$

$$A Q = 28.5 \text{ m}$$



$$\begin{aligned} \text{Final Ans} \Rightarrow AB &= AQ + QB \\ &= (28.5) \text{ m} + (1.5) \text{ m} \\ &= \underline{\underline{30 \text{ m}}} \end{aligned}$$

#LP: From the top of a building 15 m high the angle of elevation of the top of tower is found to be 30° . From the bottom of the same building, the angle of elevation of the top of the tower is found to be 60° . Find the height of the tower and the distance between the tower and the building.

Final Ans $\Rightarrow$ CT & BT

In $\triangle AQC$,
 $\tan 30 = \frac{x}{d}$
 $\frac{1}{\sqrt{3}} = \frac{x}{d}$

$d = x \cdot \sqrt{3}$

$d = \frac{15\sqrt{3}}{2} \text{ m}$

In $\triangle BTC$

$\tan 60 = \frac{x+15}{d}$

$\sqrt{3} = \frac{x+15}{d}$

$d\sqrt{3} = x+15$

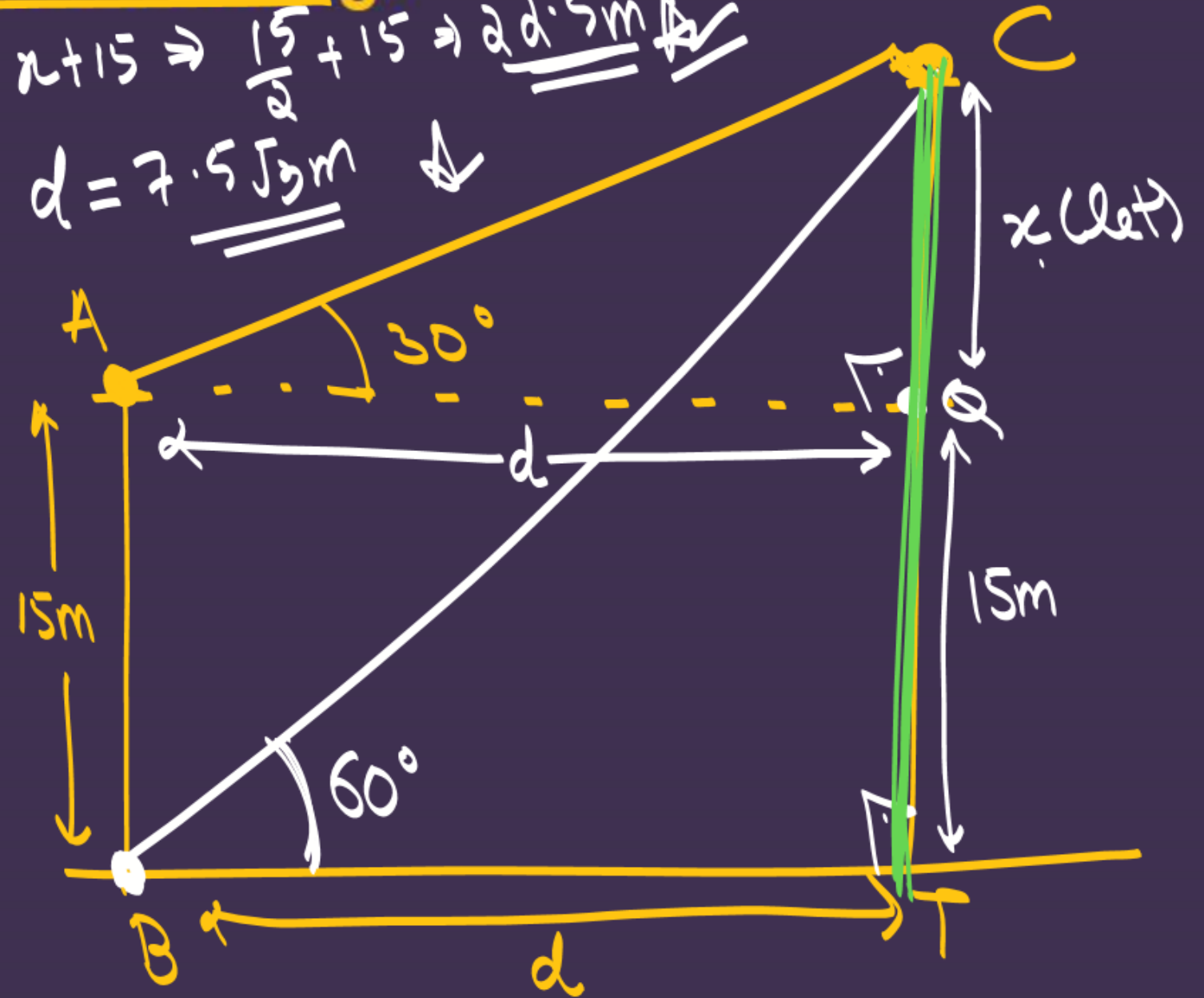
$(x\sqrt{3})\sqrt{3} = x+15$

$3x = x+15$

$2x = 15$

$x = \frac{15}{2}$

CT = $x+15 \Rightarrow \frac{15}{2} + 15 \Rightarrow \underline{\underline{2d.5 \text{ m}}}$
 BT = $d = \underline{\underline{7.5 \text{ m}}}$



#LP: Two pillars are of equal height on either sides of a road, which is 100 m wide. The angles of elevation of the top of the pillars are 60° and 30° at a point on the road between the pillars. Find the position of the point between the pillars and height of each pillar

[CBSE 2005,13,19]

Let P is at 'x' m from left tower.

In $\triangle APB$,

$$\tan 60 = \frac{H}{x}$$

$$\sqrt{3} = \frac{H}{x}$$

$$\boxed{x\sqrt{3} = H}$$

In $\triangle CPD$

$$\tan 30 = \frac{H}{100-x}$$

$$\frac{1}{\sqrt{3}} = \frac{H}{100-x}$$

$$100-x = H\sqrt{3}$$

$$\rightarrow 100-x = (x\sqrt{3})\sqrt{3}$$

$$\rightarrow 100-x = 3x$$

$$100 = 3x + x$$

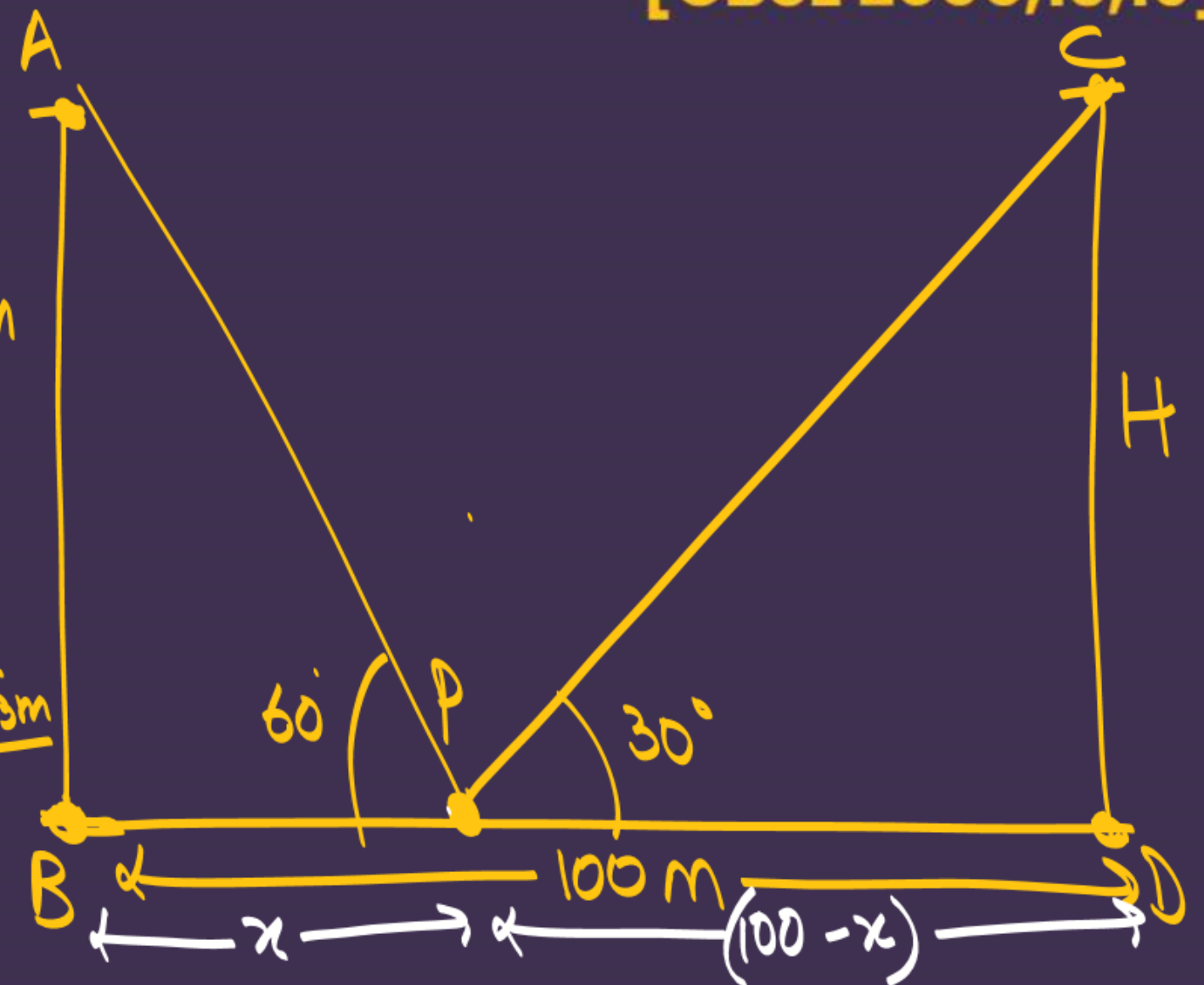
$$2500 = 4x$$

$$\boxed{x = 25m}$$

Final Ans

∴ P is 25m from left tower.

∴ height of pillar (H) = $25\sqrt{3}m$



$$\boxed{H = 25\sqrt{3}m}$$

#LP: A vertical tower stands on a horizontal plane and is surmounted by a flag staff of height 7 m. From a point on the plane, the angle of elevation of the bottom of flag staff is 30° and that of the top of the flag staff is 45° . Find the height of the tower. **[CBSE 2020]**

In $\triangle OAB$,
 $\tan 30 = \frac{H}{x}$

$$\frac{1}{\sqrt{3}} = \frac{H}{x}$$

$$x = H\sqrt{3}$$

In $\triangle OCB$
 $\tan 45 = \frac{H+7}{x}$

$$1 = \frac{H+7}{x}$$

$$x = H+7$$

$$H\sqrt{3} = H+7$$

$$H\sqrt{3} - H = 7$$

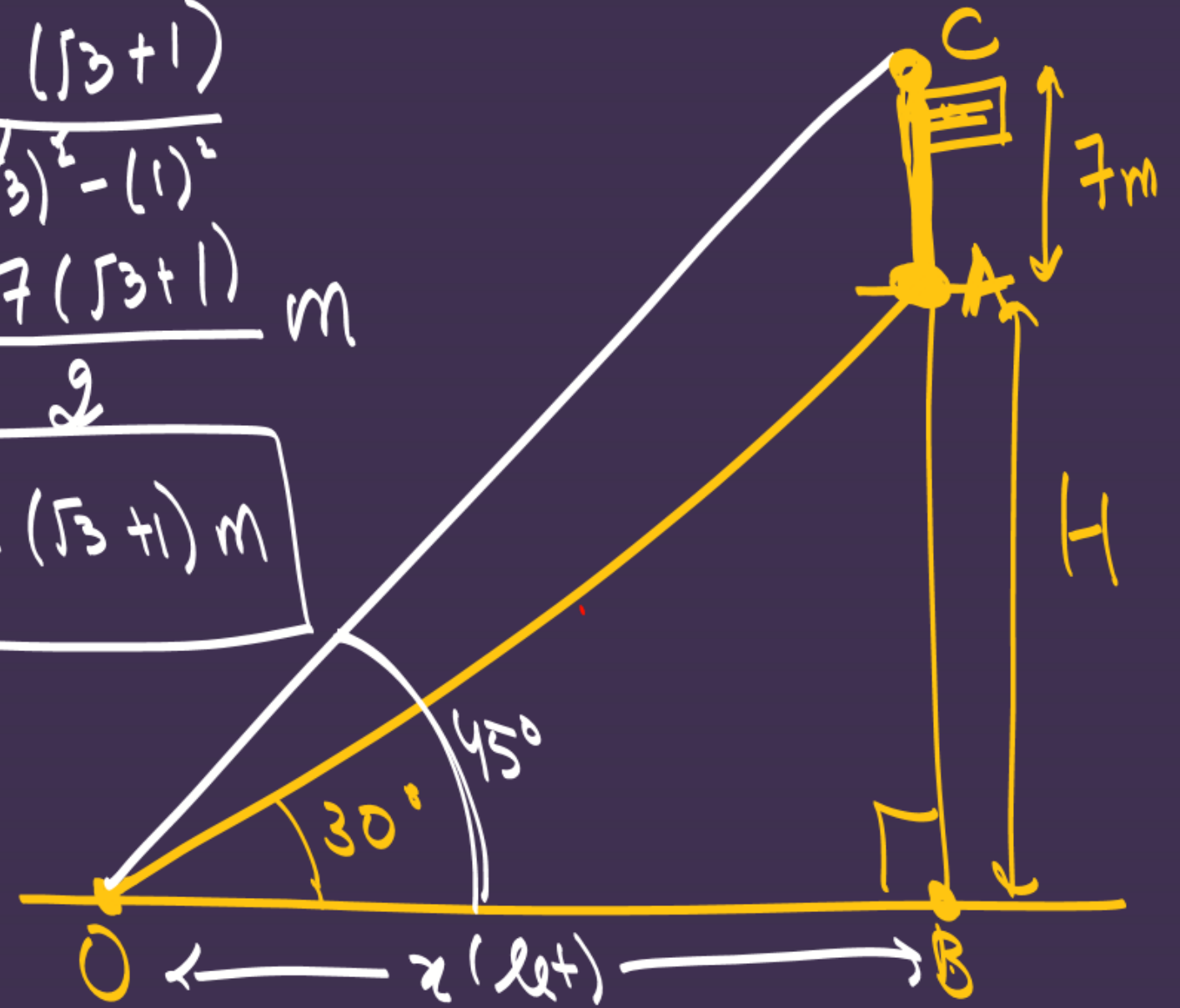
$$H(\sqrt{3} - 1) = 7$$

$$H = \frac{7}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$H = \frac{7(\sqrt{3} + 1)}{(\sqrt{3})^2 - (1)^2}$$

$$H = \frac{7(\sqrt{3} + 1)}{2} \text{ m}$$

$$H = \frac{7}{2}(\sqrt{3} + 1) \text{ m}$$



#LP: A vertical tower stands on a horizontal plane and is surmounted by a vertical flag staff of height h .

At a point on the plane, the angles of Elevation of the bottom and the top of the flag staff are

α and β respectively. Prove that the height of the tower is

$$\frac{h \tan \alpha}{\tan \beta - \tan \alpha}$$

In $\triangle OCB$,

$$\tan \alpha = \frac{H}{x}$$

$$x = \frac{H}{\tan \alpha}$$

In $\triangle OAB$

$$\tan \beta = \frac{H + h}{x}$$

$$x \times \tan \beta = H + h$$

$$\frac{H}{\tan \alpha} \tan \beta = H + h$$

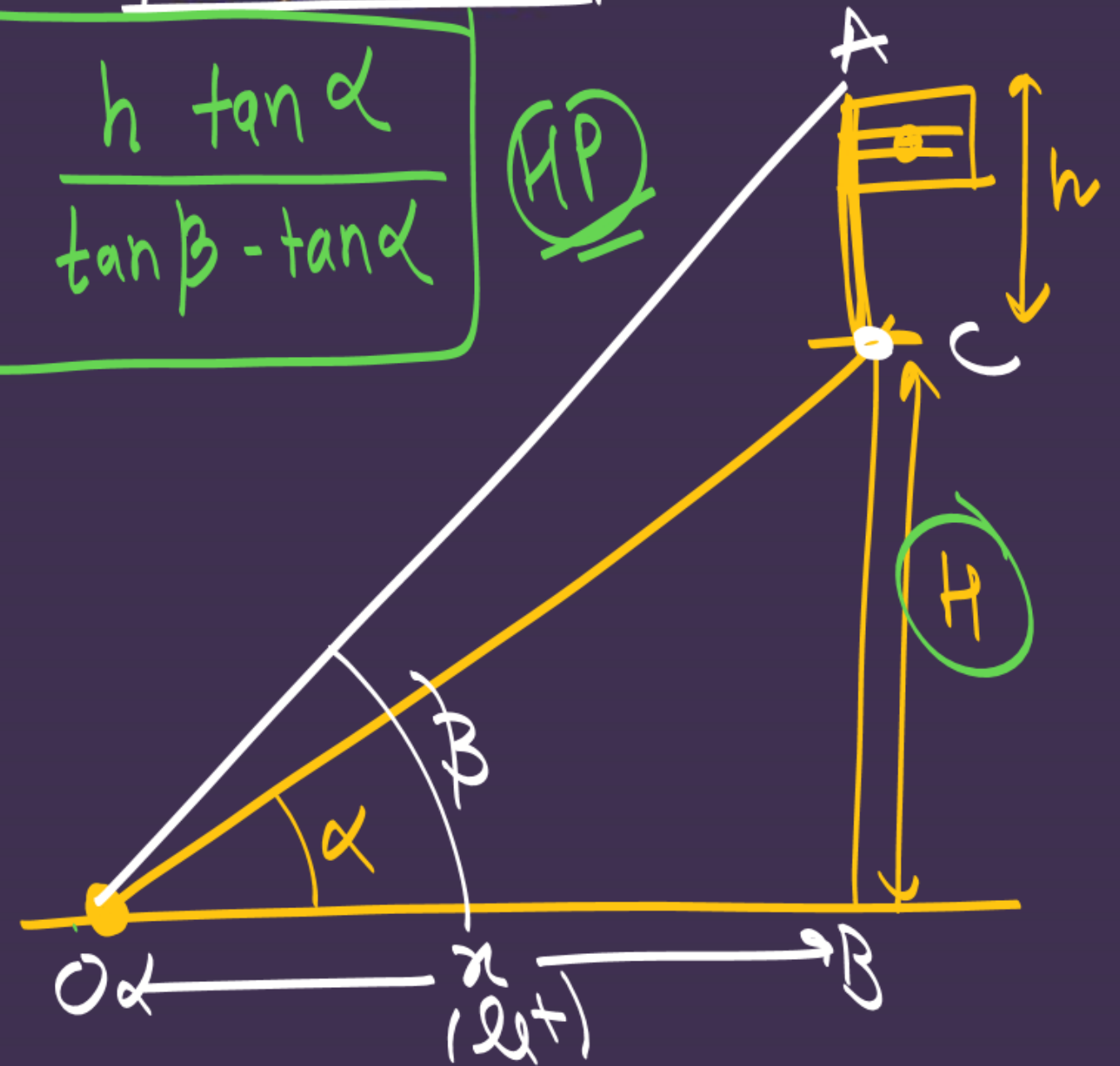
$$H \frac{\tan \beta}{\tan \alpha} - H = h$$

$$H \left(\frac{\tan \beta}{\tan \alpha} - 1 \right) = h$$

$$H \left(\frac{\tan \beta - \tan \alpha}{\tan \alpha} \right) = h$$

$$H = \frac{h \tan \alpha}{\tan \beta - \tan \alpha}$$

(HP)



#LP: On the same side of a tower, two objects are located. When observed from the top of the tower, their angles of depression are 45° and 60° . If the height of the tower is 150 m, find the distance between the objects.

In $\triangle OBC$

$$\tan 60 = \frac{150}{x}$$

$$\sqrt{3} = \frac{150}{x}$$

$$x = \frac{150}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$x = \frac{50}{1} \sqrt{3}$$

$$\boxed{x = 50\sqrt{3} \text{ m}}$$

In $\triangle OAC$,

$$\tan 45 = \frac{150}{d+x}$$

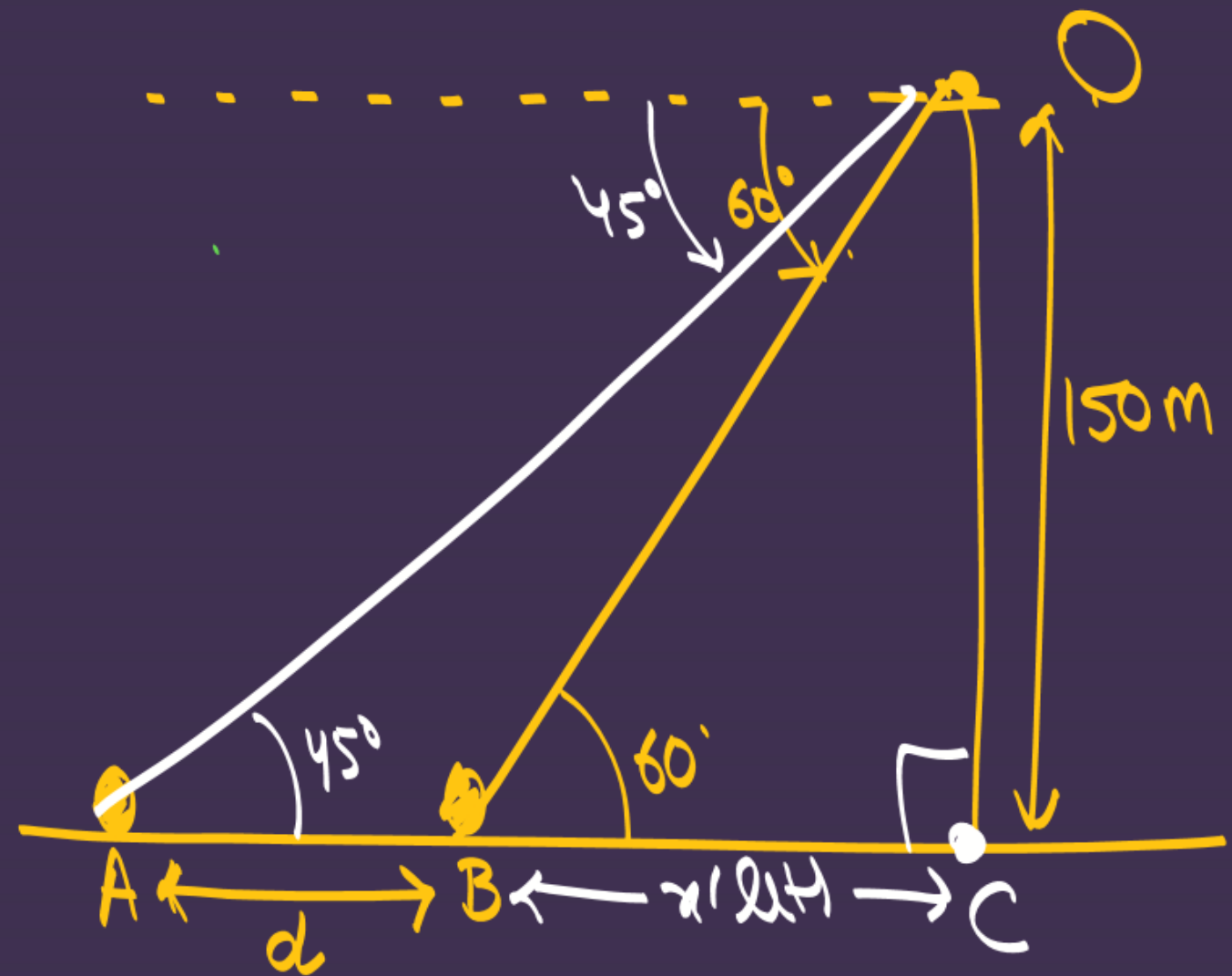
$$1 = \frac{150}{d+x}$$

$$d+x = 150$$

$$d + 50\sqrt{3} = 150$$

$$d = 150 - 50\sqrt{3}$$

$$\boxed{d = 50(3 - \sqrt{3}) \text{ m}}$$



#LP: The angles of depression of two ships from the top of a light house and on the same side of it are found to be 45° and 30°. If the ships are 200 m apart, find the height of the light house.

In $\triangle OCB$,

$$\tan 45 = \frac{H}{x}$$

$$1 = \frac{H}{x}$$

$$\boxed{x = H}$$

In $\triangle OAC$

$$\tan 30 = \frac{H}{200 + x}$$

$$\frac{1}{\sqrt{3}} = \frac{H}{200 + x}$$

$$200 + x = H\sqrt{3}$$

$$200 + H = H\sqrt{3}$$

$$200 = H\sqrt{3} - H$$

$$200 = H(\sqrt{3} - 1)$$

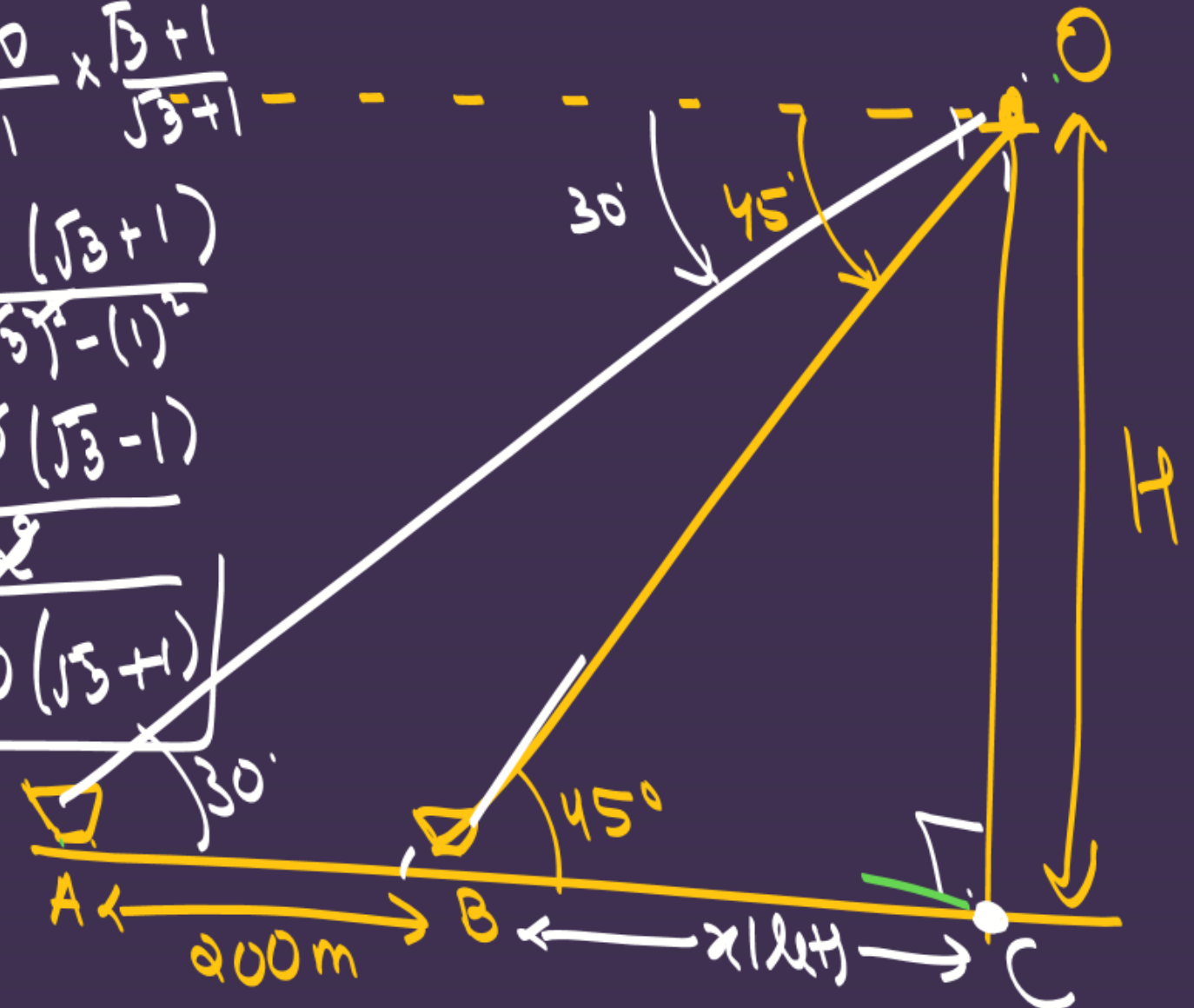
$$\frac{200}{\sqrt{3} - 1} = H$$

$$H = \frac{200}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$H = \frac{200(\sqrt{3} + 1)}{(\sqrt{3})^2 - (1)^2}$$

$$H = \frac{200(\sqrt{3} + 1)}{2}$$

$$\boxed{H = 100(\sqrt{3} + 1)}$$



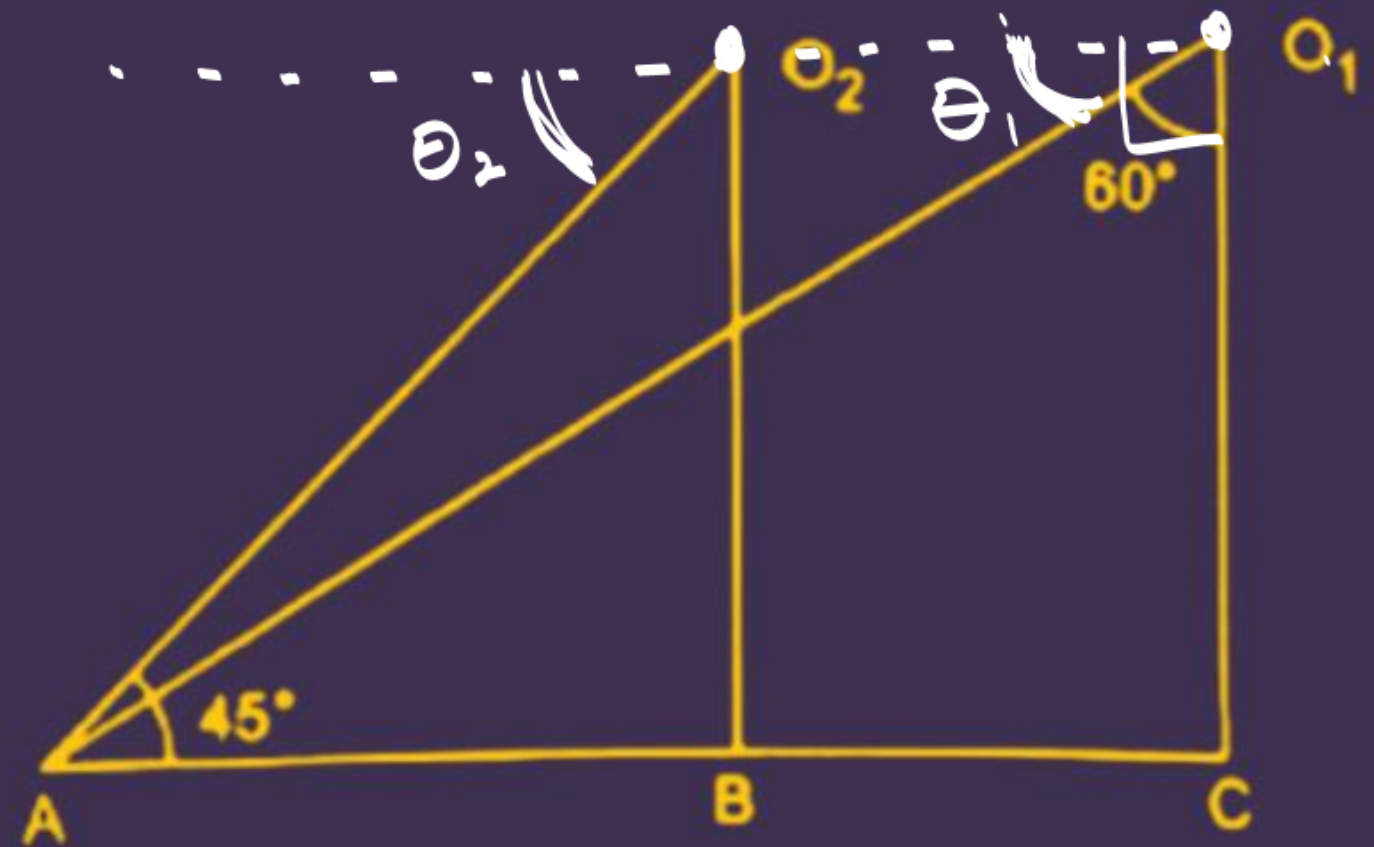
#LP: In the given figure, find the angles of depressions from the observing positions O_1 and O_2 respectively of the object A.

$$\boxed{\theta_2 = 45^\circ} \text{ (A.I.A)}$$

$$\theta_1 + 60 = 90$$

$$\theta_1 = 90 - 60$$

$$\boxed{\theta_1 = 30^\circ} \checkmark$$



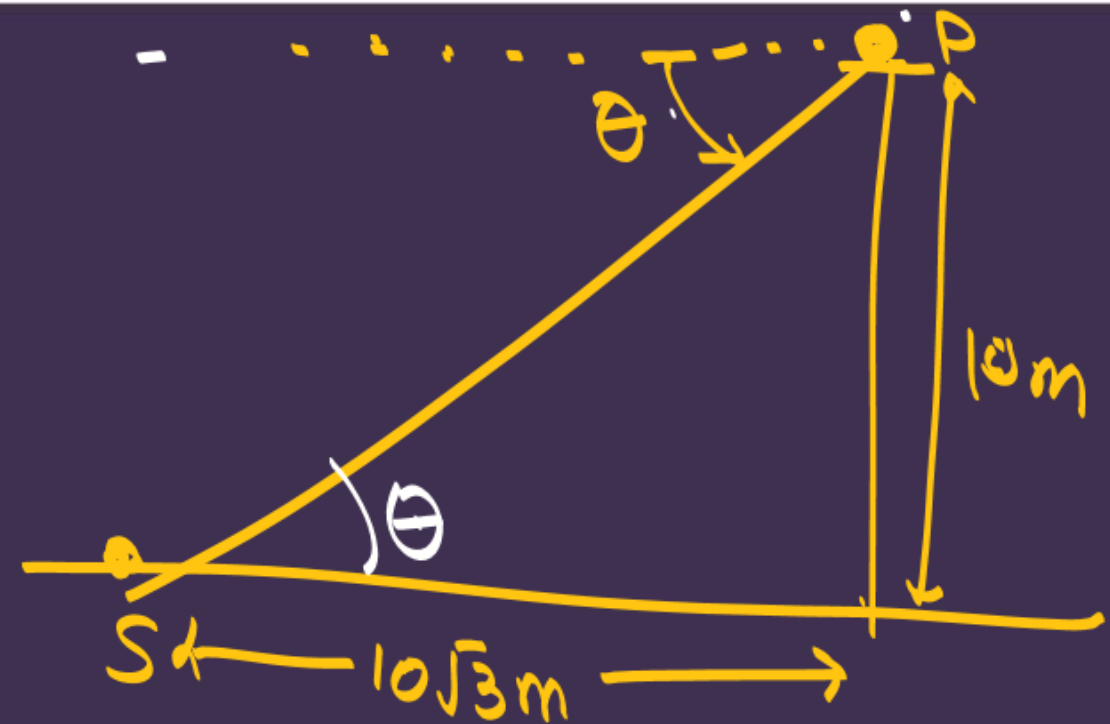
#LP: A peacock sitting on the top of a tree of height 10 m observes a snake moving on the ground. If the snake is $10\sqrt{3}$ m away from the base of the tree, then angle of depression of the snake from the eye of the peacock is **[CBSE 2025]**

- ☒ (a) 30°
- (b) 45°
- (c) 60°
- (d) 90°

$$\begin{aligned}\tan \theta &= \frac{P}{B} \\ \tan \theta &= \frac{10}{10\sqrt{3}} \\ \tan \theta &= \frac{1}{\sqrt{3}}\end{aligned}$$

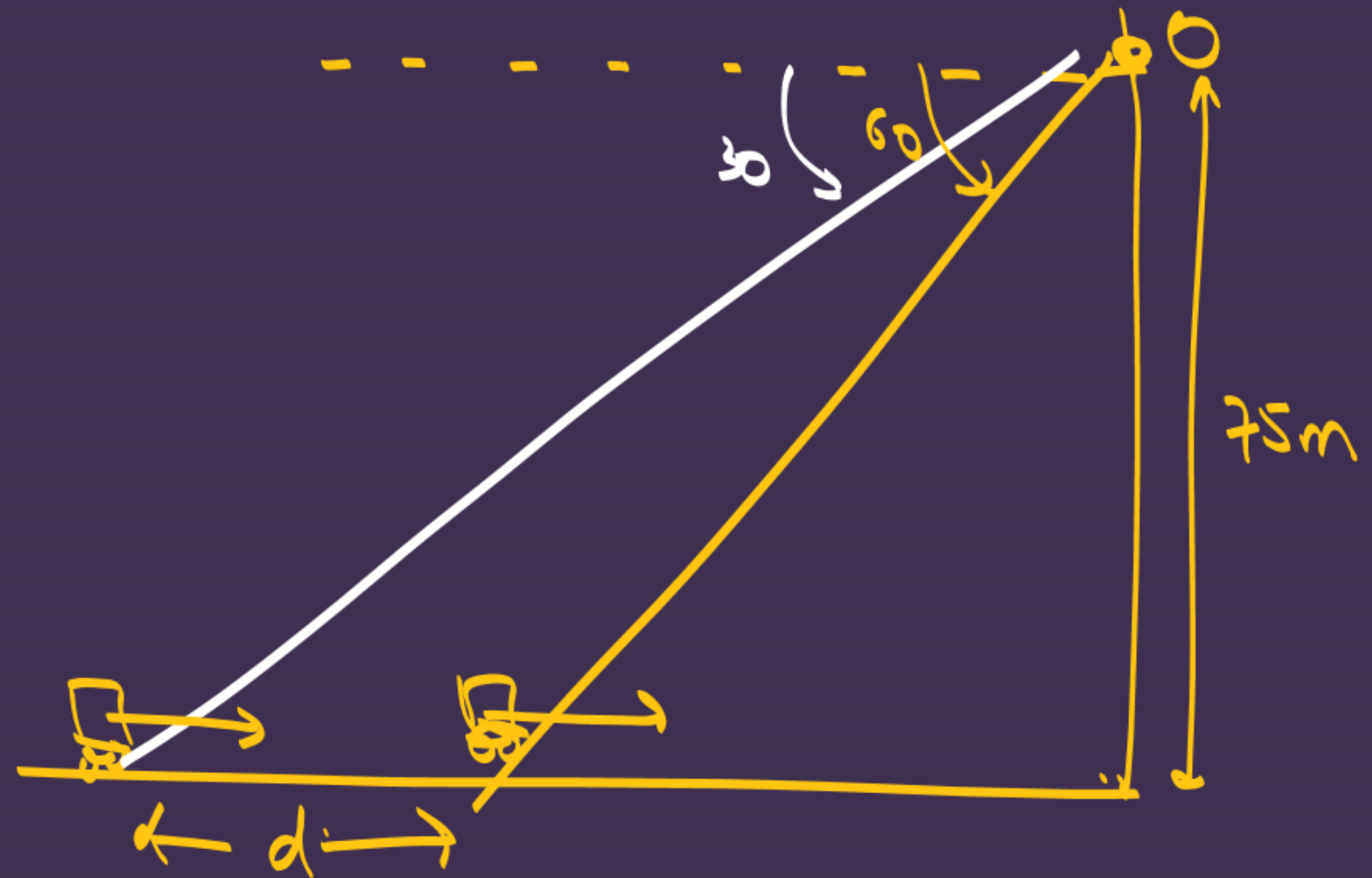
comparing

$$\boxed{\theta = 30^\circ}$$
$$\boxed{\tan \theta = \tan 30^\circ}$$



#LP: A straight highway leads to the foot of a tower. A man standing on the top of the 75 m high observes two cars at angles of depression of 30° and 60° which are approaching the foot of the tower. If one car is exactly behind the other on the same side of the tower, find the distance between the two cars. (Use $\sqrt{3} = 1.73$) (CBSE 2023) , [CBSE 2008,09,17]

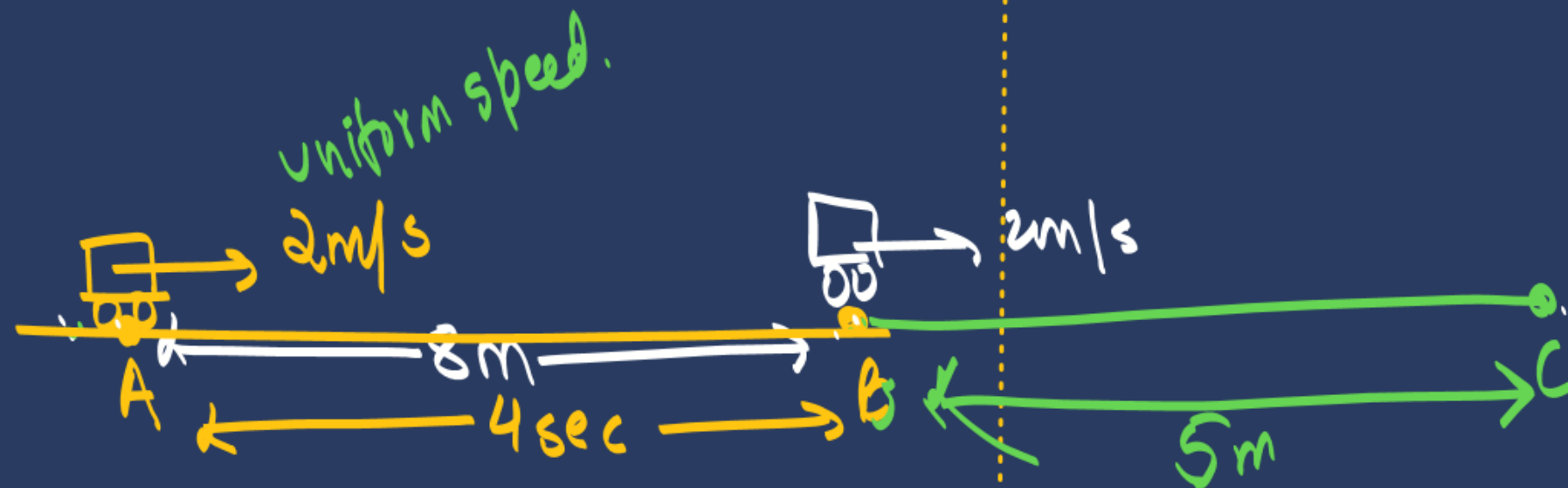
HW



#(K3B)

$$d = s \times t$$

uniform speed/constant speed → same speed



$$\begin{aligned} d &= s \times t \\ AB &= s \times t \\ &= 2 \times 4 \end{aligned}$$

$$\boxed{AB = 8m}$$

Find the time taken by car to go from B to C?

$$\begin{aligned} d &= s \times t \\ BC &= 2 \times t \\ s &= 2 \times t \end{aligned}$$

$$t = \frac{s}{2} = 2.5 \text{ sec}$$

#LP: A man rowing a boat away from a lighthouse 150 m high takes 2 minutes to change the angle of elevation of the top of lighthouse from 45° to 30° . Find the speed of the boat. (Use $\sqrt{3} = 1.732$)

$$d = s \times t$$

$$\text{IF} = s \times (120)$$

$$150(\sqrt{3}-1) = s \times 120$$

$$\frac{5}{4}(\sqrt{3}-1) \text{ m/s} = s$$

In $\triangle OIQ$

$$\tan 45 = \frac{150}{x}$$

$$1 = \frac{150}{x}$$

$$x = 150 \text{ m}$$

In $\triangle OFQ$

$$\tan 30 = \frac{150}{x+d}$$

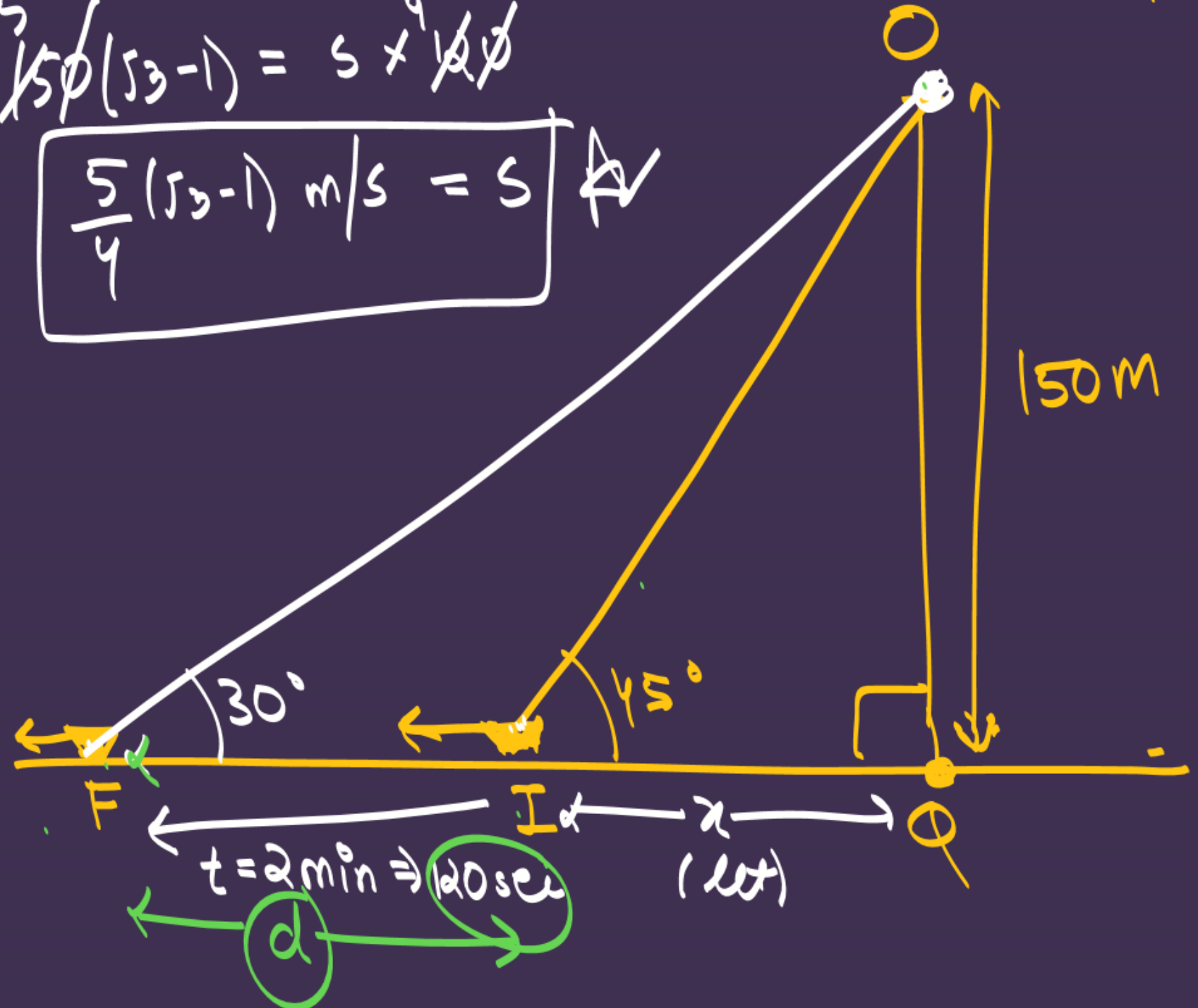
$$\frac{1}{\sqrt{3}} = \frac{150}{x+d}$$

$$x+d = 150\sqrt{3}$$

$$150 + d = 150\sqrt{3}$$

$$d = 150\sqrt{3} - 150$$

$$d = 150(\sqrt{3}-1) \text{ m}$$

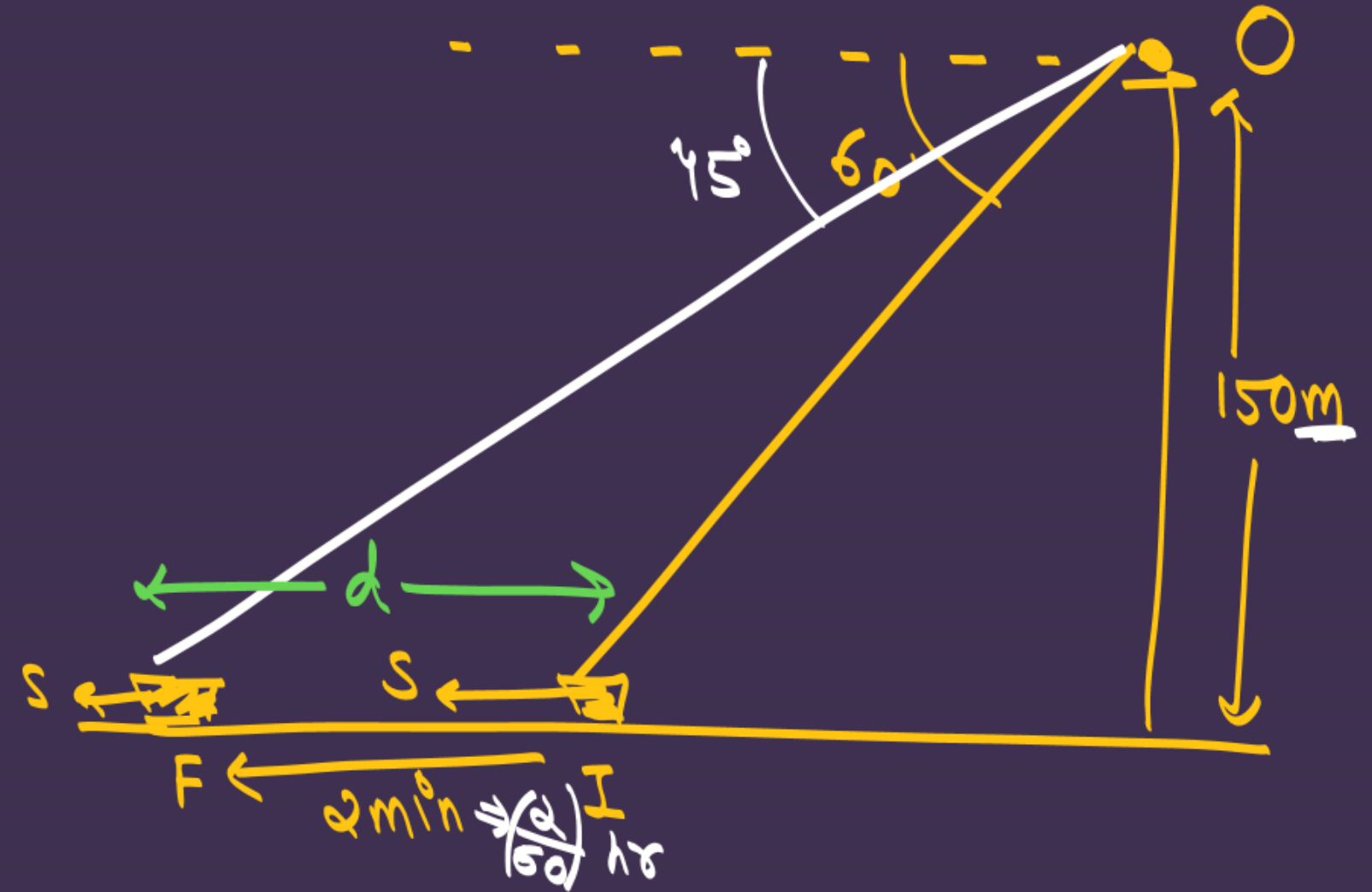


#LP: A moving boat is observed from the top of a 150 m high cliff moving away from the cliff. The angle of depression of the boat changes from 60° to 45° in 2 minutes. Find the speed of the boat in m/h.

[CBSE 2017]

$$d = s \times t$$

$$d = ? \quad \text{IF} = s \times \left(\frac{2}{60}\right) \text{ hr}$$



#LP: A Ladder set against a wall at an angle 45° to the ground. If the foot of the ladder is pulled away from the wall through a distance of 4 m, its top slides a distance of 3 m down the wall making an angle 30° with the ground. Find the final height of the top of the ladder from the ground and length of the ladder. (2023)

$$AB = A'B' = L$$

In $\triangle A'B'O$

$$\tan 30 = \frac{H}{x+4}$$

$$\frac{1}{\sqrt{3}} = \frac{H}{x+4}$$

$$x+4 = H\sqrt{3}$$

$$H+3+4 = H\sqrt{3}$$

$$H+7 = H\sqrt{3}$$

In $\triangle ABO$

$$\tan 45 = \frac{H+3}{x}$$

$$1 = \frac{H+3}{x}$$

$$x = H+3$$

$$x = H(\sqrt{3}-1)$$

$$H = \frac{7}{\sqrt{3}-1} \text{ m}$$



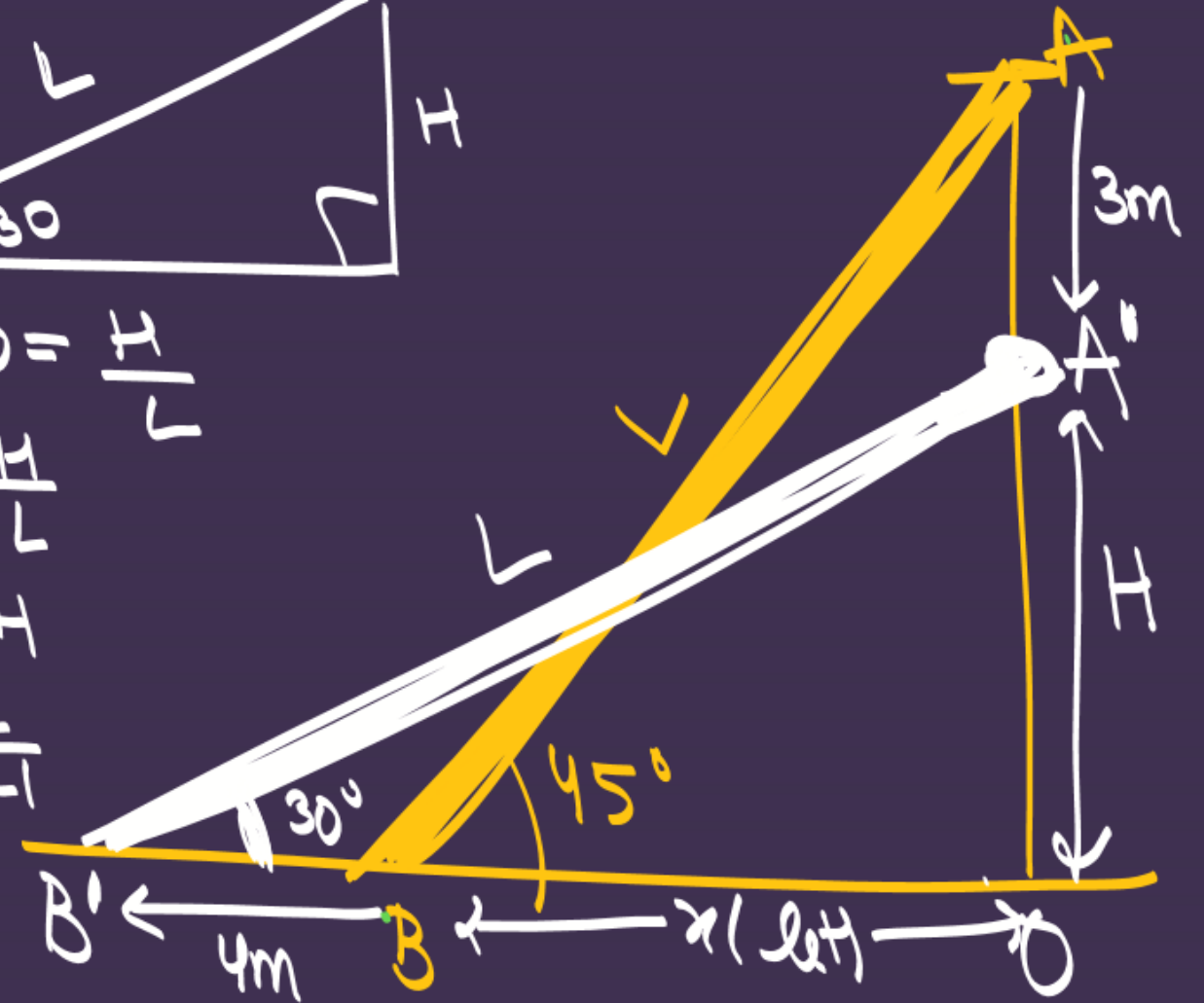
$$\sin 30 = \frac{H}{L}$$

$$\frac{1}{2} = \frac{H}{L}$$

$$L = 2H$$

$$L = 2x \frac{7}{\sqrt{3}-1}$$

$$L = \frac{14}{\sqrt{3}-1} \text{ m}$$



#LP: The angle of elevation of the top of the building from the foot of the tower is 30° and the angle of elevation of the top of the tower from the foot of the building is 60° . If the tower is 50 m high, find the height of the building.

In $\triangle ATB$,

$$\tan 60 = \frac{50}{x}$$

$$\sqrt{3} = \frac{50}{x}$$

$$x = \frac{50}{\sqrt{3}}$$

$\triangle CTB$,

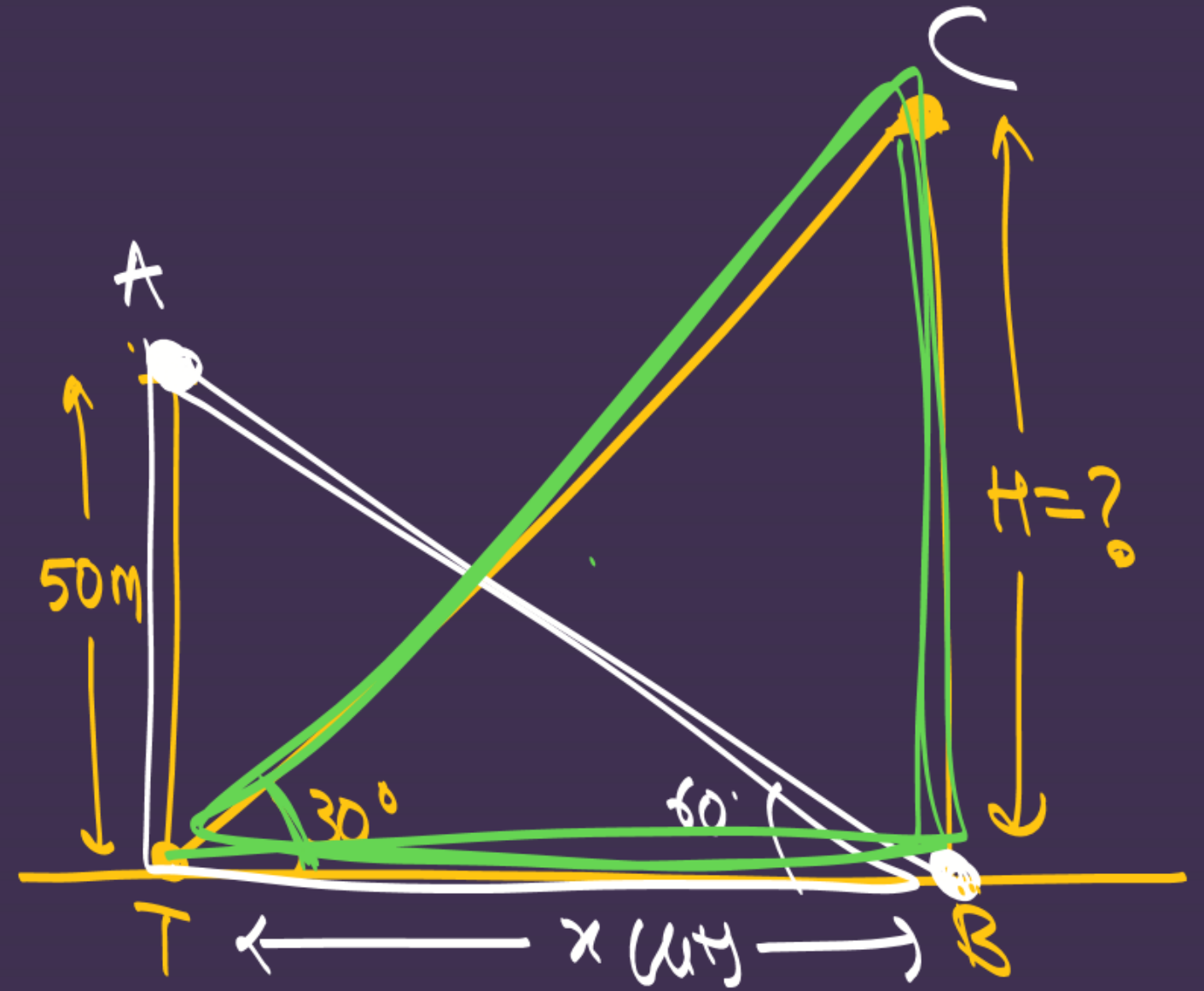
$$\tan 30 = \frac{H}{x}$$

$$\frac{1}{\sqrt{3}} = \frac{H}{x}$$

$$\frac{1}{\sqrt{3}} \times x = H$$

$$\frac{1}{\sqrt{3}} \times \frac{50}{\sqrt{3}} = H$$

$$H = \frac{50}{3}$$



#LP: Amit, standing on a horizontal plane, finds a bird flying at a distance of 200 m from him at an elevation of 30° . Deepak standing on the roof of a 50 m high building. finds the angle of elevation of the same bird to be 45° . Amit and Deepak are on opposite sides of the bird. Find the distance of the bird from Deepak. (2019)

Conc $\rightarrow OB \perp AC$

In $\triangle OBA$

$$\sin 30 = \frac{OB}{200}$$

$$\frac{1}{2} = \frac{OB}{200}$$

$$OB = 100 \text{ m}$$

$$QB = 100 - 50$$

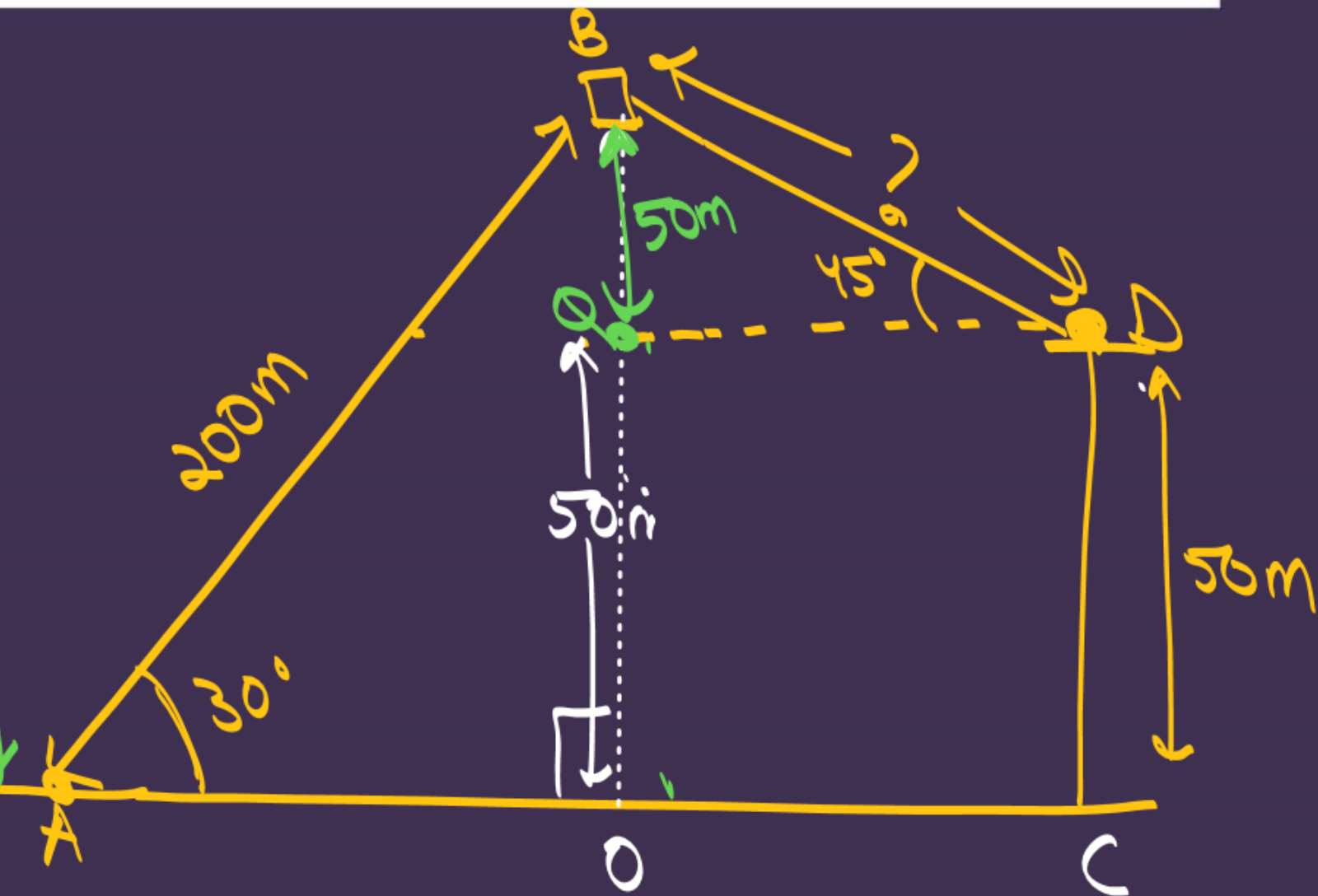
$$QB = 50 \text{ m}$$



$$\sin 45 = \frac{QB}{BD}$$

$$\frac{1}{\sqrt{2}} = \frac{50}{BD}$$

$$BD = 50\sqrt{2} \text{ m}$$



#LP: A man standing on the deck of a ship, which is 10 m above water level, observes the angle of elevation of the top of a hill as 60° and the angle of depression of the base of a hill as 30° . Find the distance of the hill from the ship and the height of the hill.

[CBSE 2016]

In $\triangle OQB$

$$\tan 30 = \frac{10}{d}$$

$$\frac{1}{\sqrt{3}} = \frac{10}{d}$$

$$d = 10\sqrt{3}\text{m}$$

In $\triangle OAQ$

$$\tan 60 = \frac{AQ}{d}$$

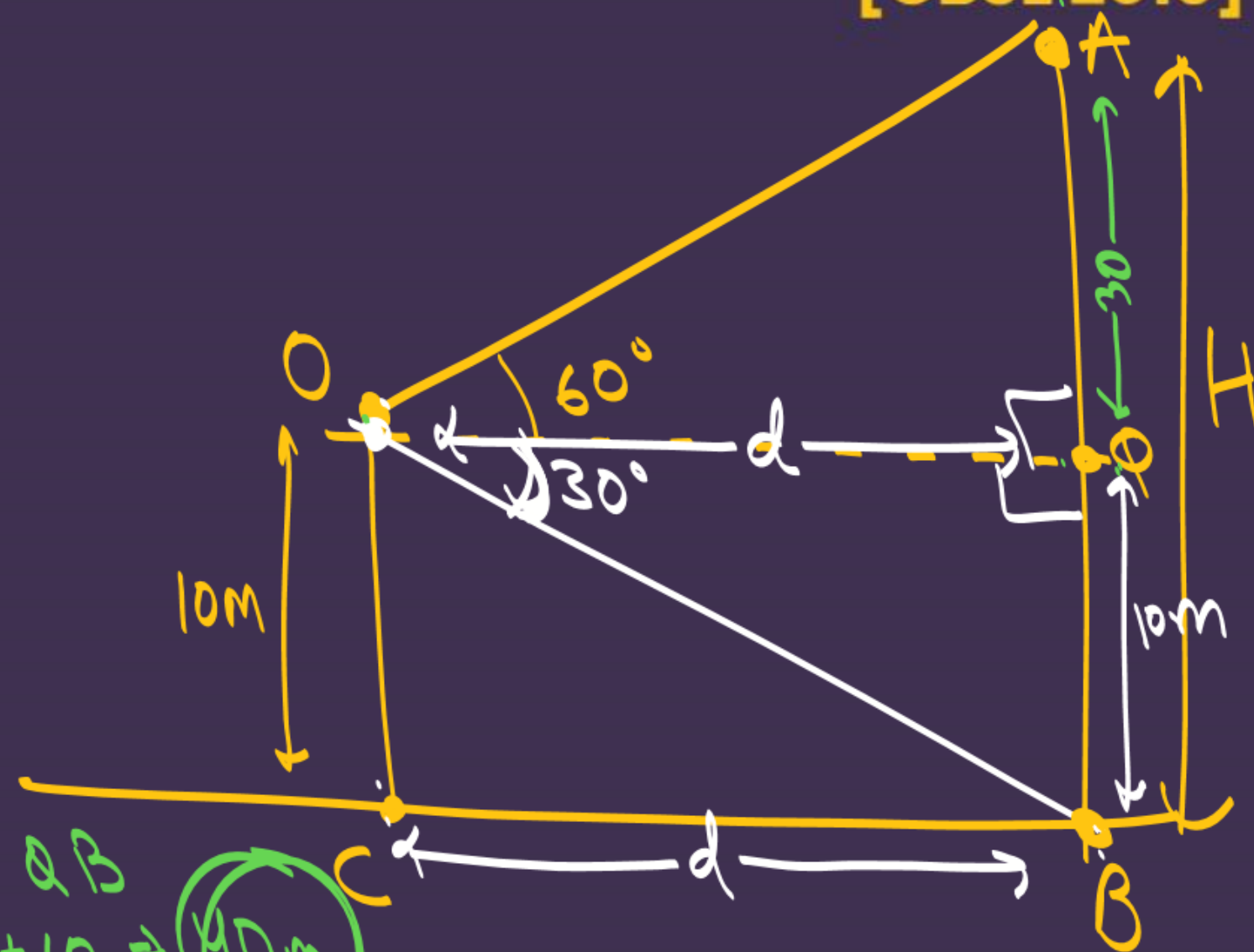
$$\sqrt{3} = \frac{AQ}{d}$$

$$\sqrt{3} \times d = AQ$$

$$\sqrt{3} \times 10\sqrt{3} = AQ$$

$$30 = AQ$$

$$\begin{aligned} \text{Final Ans} \Rightarrow H &= AQ + QB \\ &= 30 + 10 = 40\text{m} \end{aligned}$$



#LP: The angle of elevation of a cloud from a point 60m above a lake is 30° and the angle of depression of its reflection in the lake is 60° . Find the height of the cloud. **[CBSE 2010,17]**

In $\triangle AOD$

$$\tan 30 = \frac{AD}{d}$$

$$\frac{1}{\sqrt{3}} = \frac{AD}{d}$$

$$d = AD \sqrt{3}$$

$$d = (H - 60) \sqrt{3}$$

In $\triangle ODB$

$$\tan 60 = \frac{H + 60}{d}$$

$$\sqrt{3} = \frac{H + 60}{d}$$

$$d \sqrt{3} = H + 60$$

$$[(H - 60) \sqrt{3}] \sqrt{3} = H + 60$$

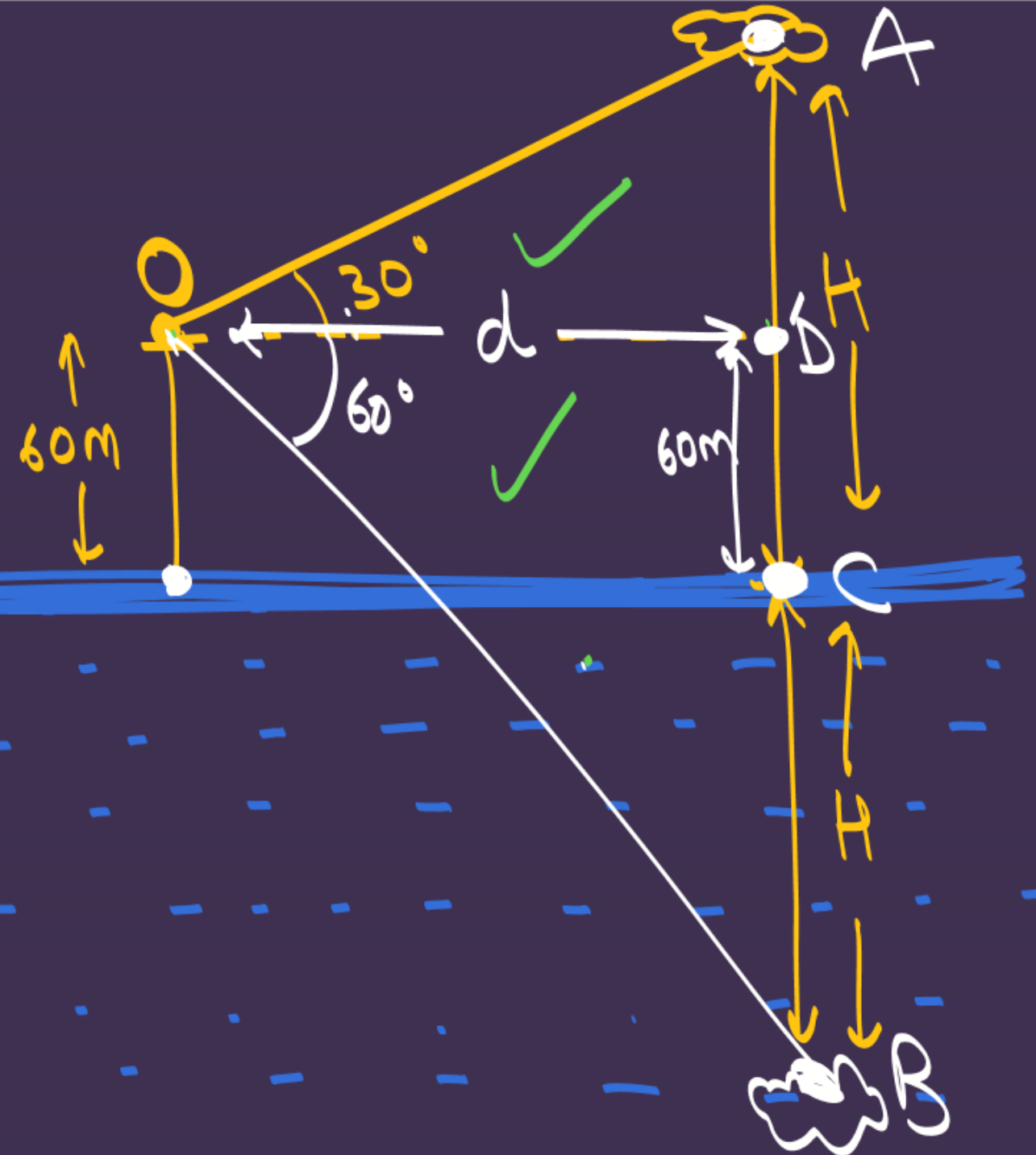
$$(H - 60) 3 = H + 60$$

$$3H - 180 = H + 60$$

$$3H - H = 180 + 60$$

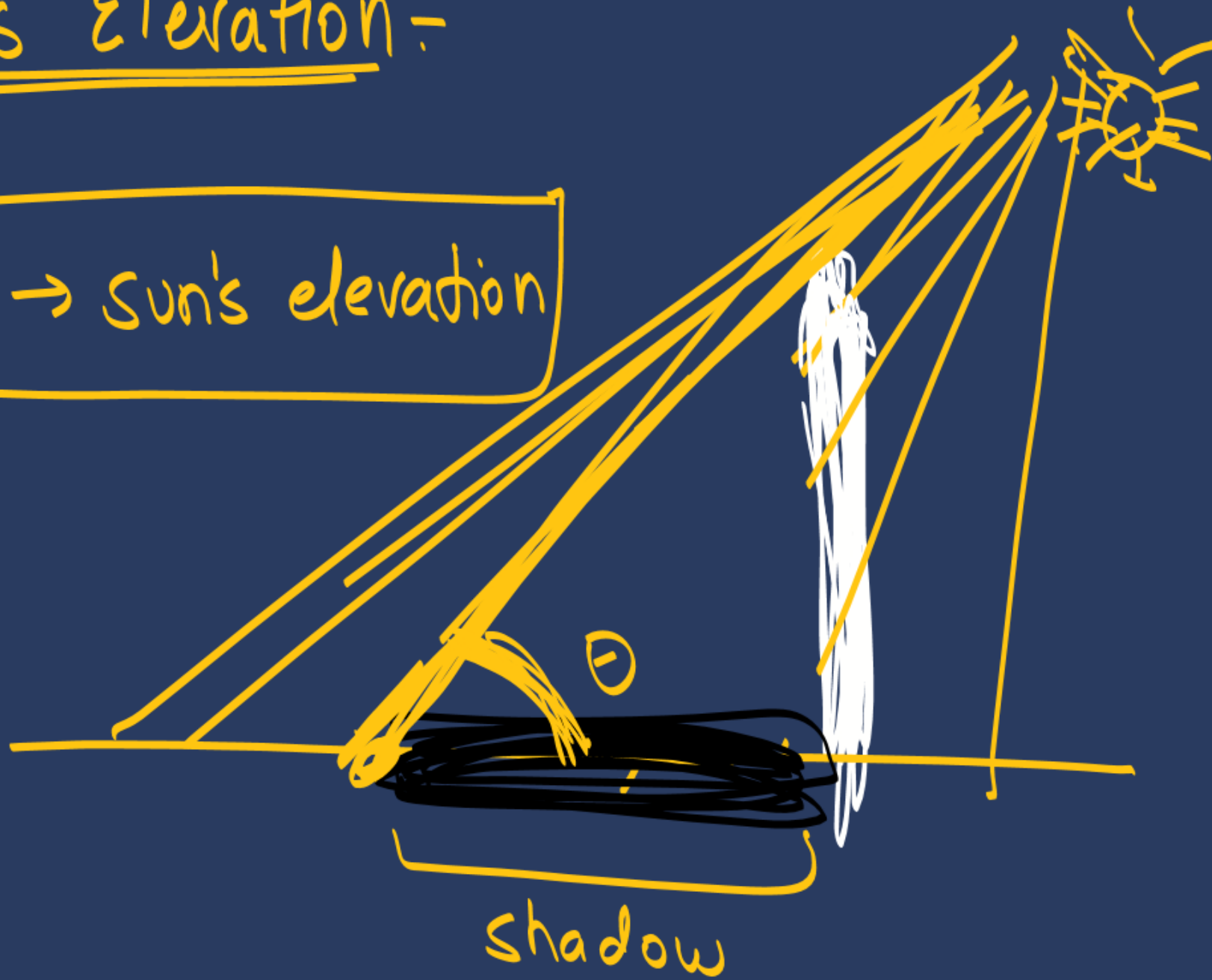
$$2H = 240$$

$$H = 120m$$

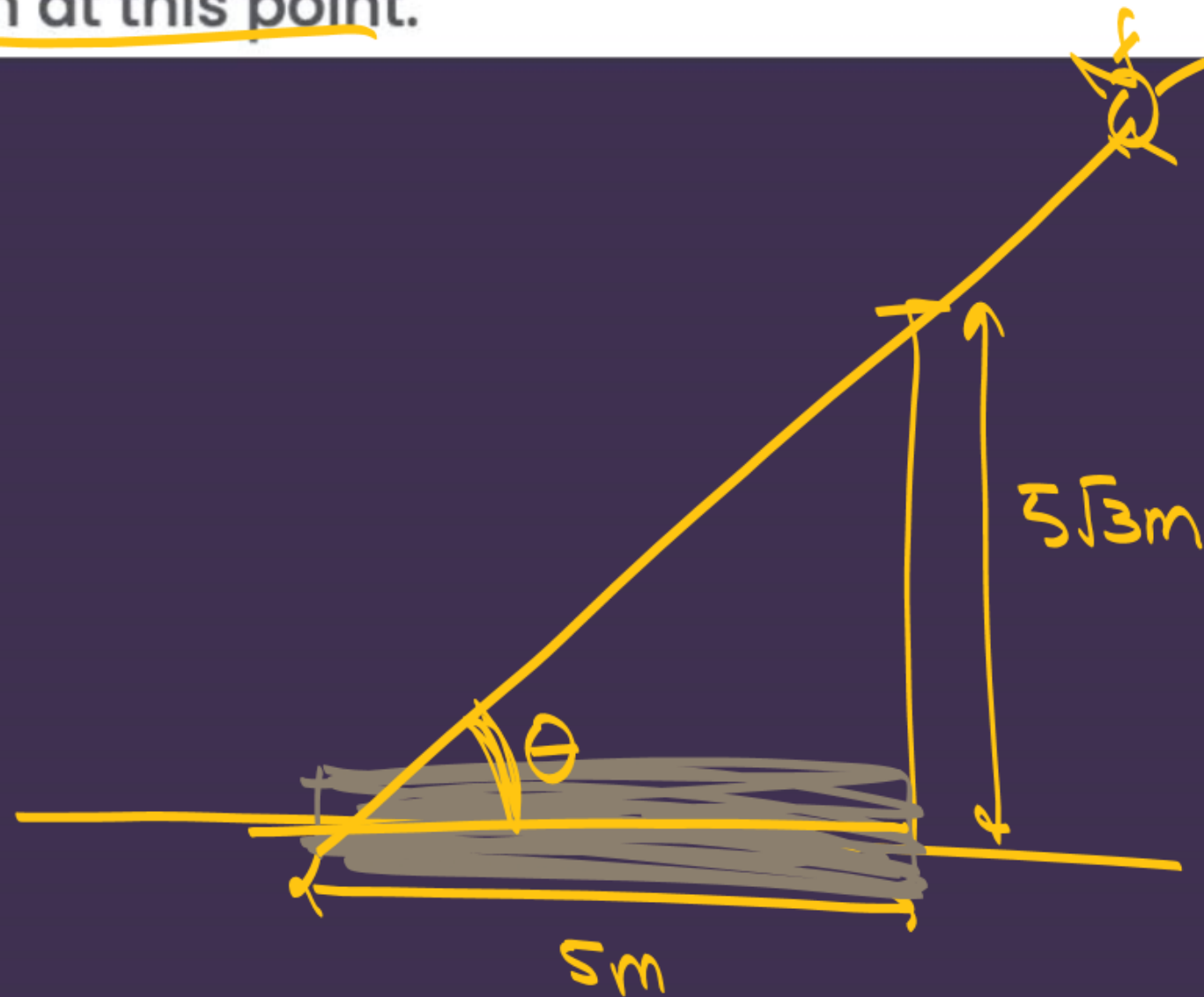


Sun's Elevation:-

$\theta \rightarrow$ sun's elevation



#LP: A lamp post, $5\sqrt{3}$ m high, cast a shadow 5 m long on the ground. Find the sun's elevation at this point.



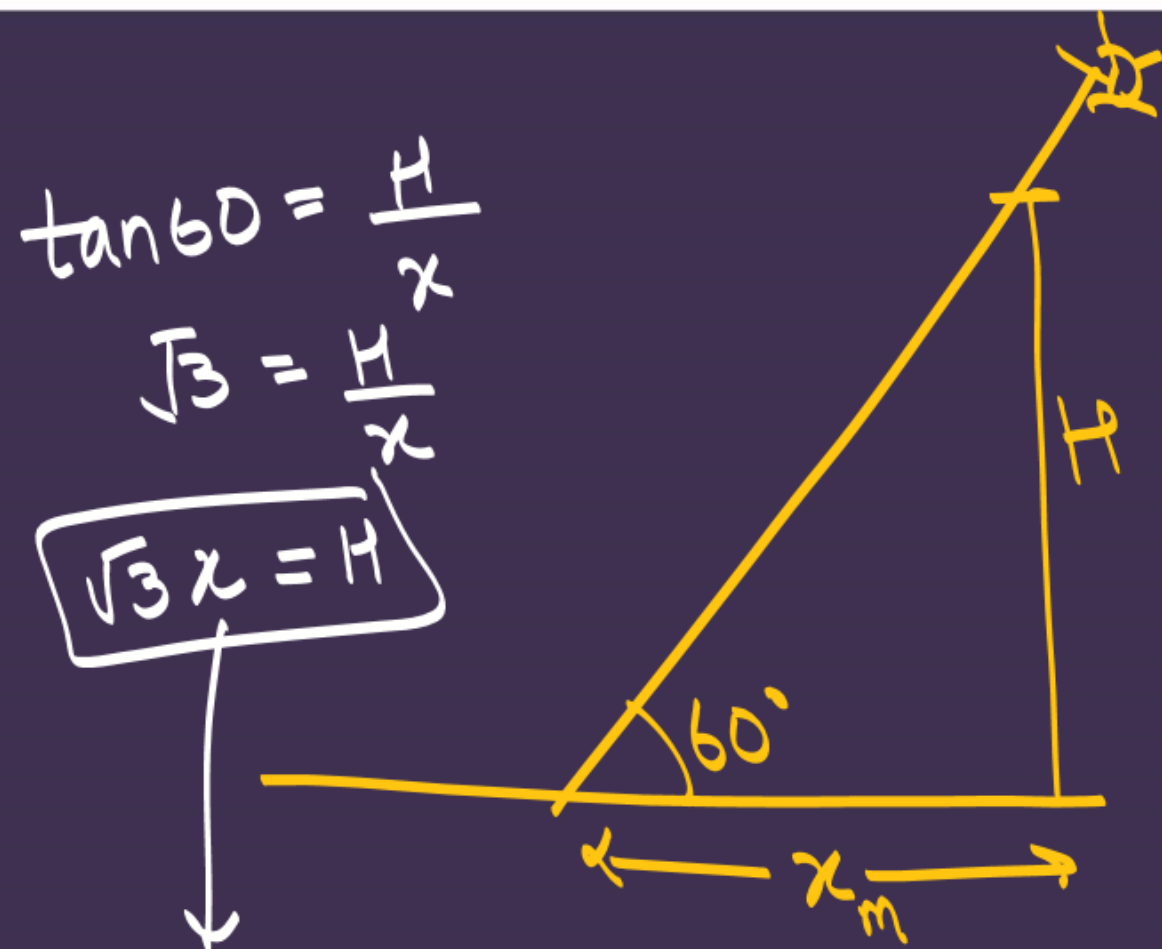
$$\tan \theta = \frac{5\sqrt{3}}{5}$$

$$\tan \theta = \sqrt{3}$$

$$\tan \theta = \tan 60$$

comparing $\Rightarrow \boxed{\theta = 60^\circ}$

#LP: The shadow of a tower standing on a level ground is found to be 40 m longer when the Sun's altitude is 30° than when it is 60° . Find the height of the tower. (2022)



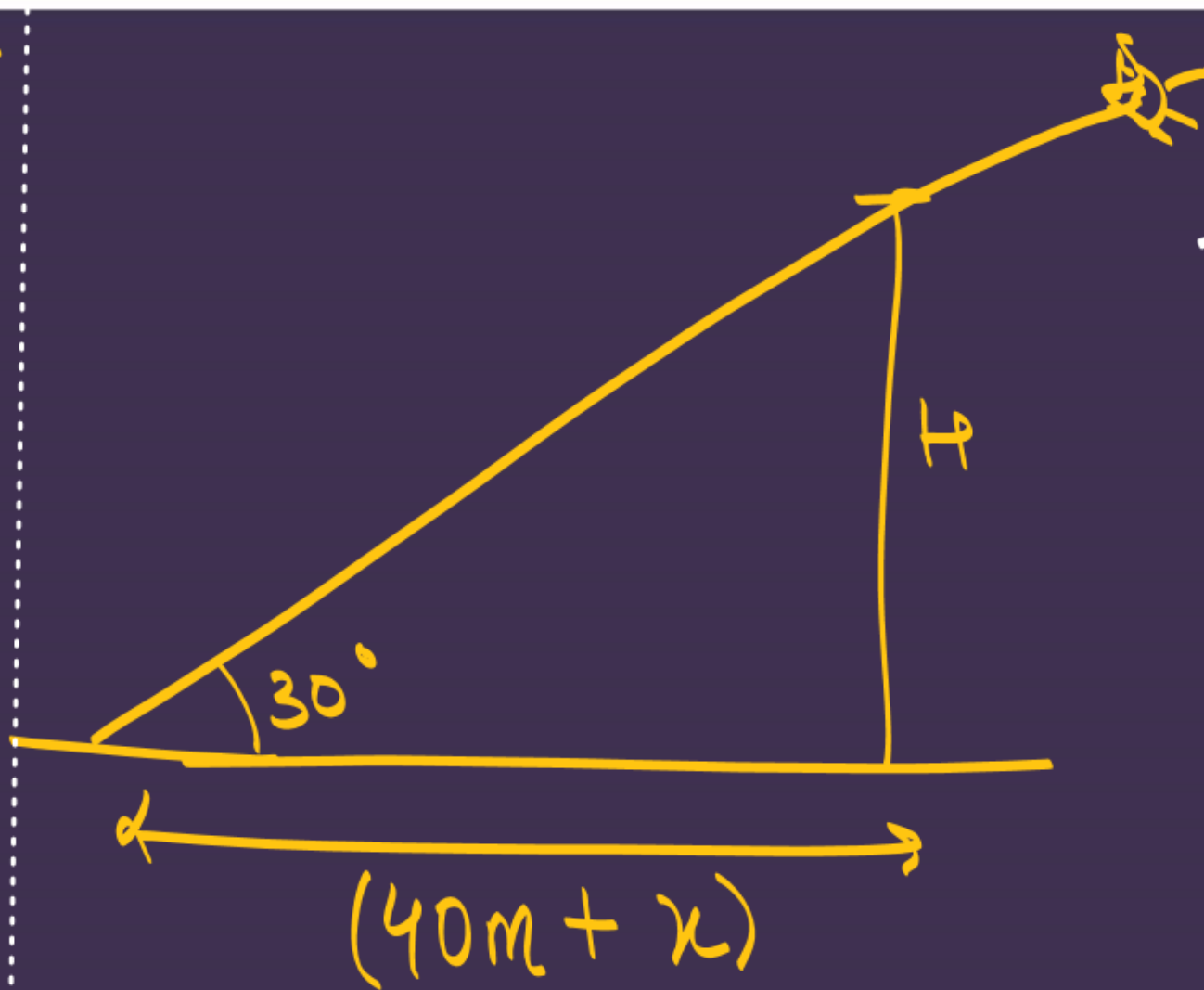
$$\tan 60 = \frac{H}{x}$$

$$\sqrt{3} = \frac{H}{x}$$

$$\sqrt{3}x = H$$

$$H = \sqrt{3}(20)$$

$$H = 20\sqrt{3} \text{ m} \quad \text{Ans}$$



$$\tan 30 = \frac{H}{40+x}$$

$$\frac{1}{\sqrt{3}} = \frac{H}{40+x}$$

$$40+x = H\sqrt{3}$$

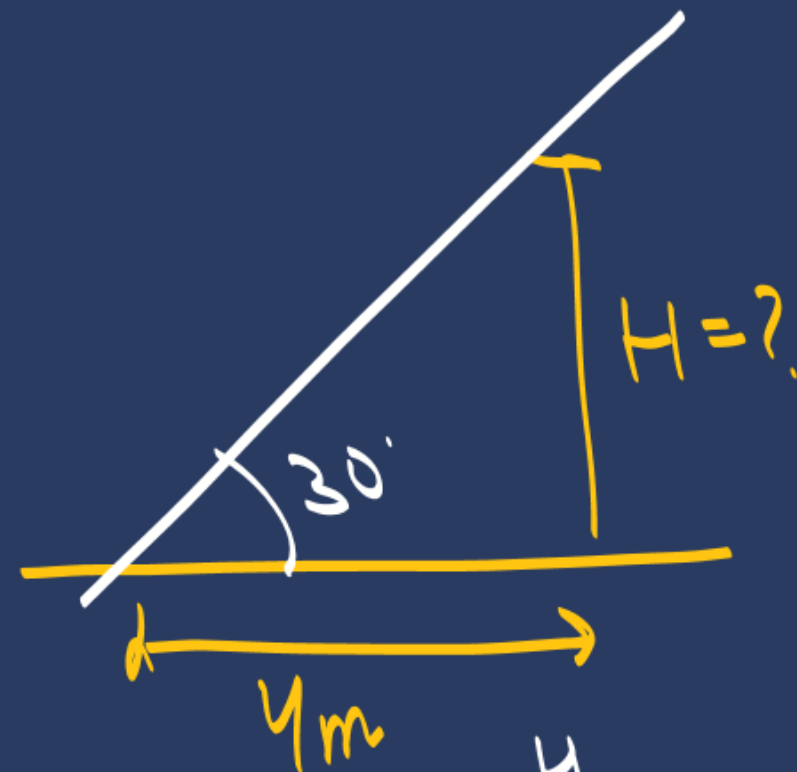
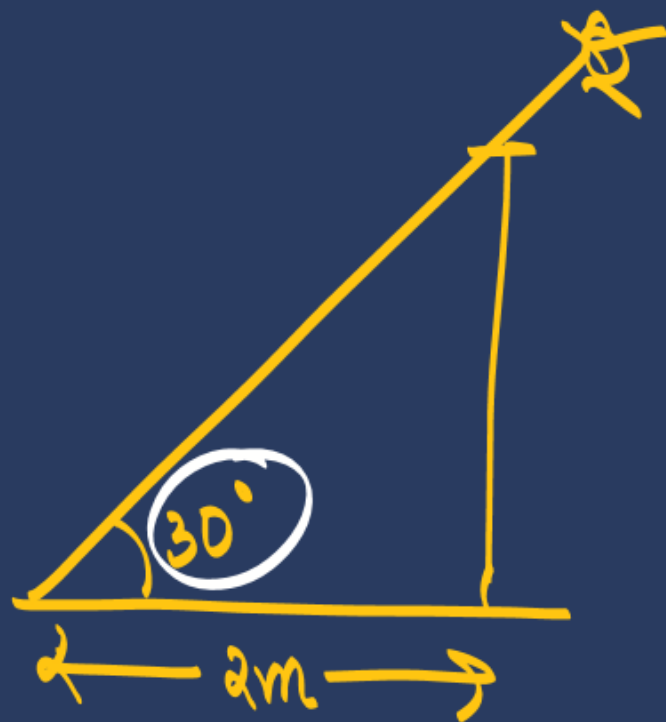
$$40+x = (\sqrt{3}x)\sqrt{3}$$

$$40+x = 3x$$

$$20 = 2x$$

$$x = 20$$

"same time" $\rightarrow \theta \rightarrow$ same



$$\tan 30 = \frac{H}{4}$$

$$\frac{1}{\sqrt{3}} = \frac{H}{4}$$
$$H = \frac{4}{\sqrt{3}} \text{ m}$$

#LP: The shadow of a flagstaff is three times as long as the shadow of the flagstaff when the sunrays meet the ground at an angle of 60° . Find the angle between the sunrays and the ground at the time of longer shadow.

#LP:

A round balloon of radius r subtends an angle α at the eye of the observer while the angle of elevation of its centre is β . Prove that the height of the centre of the balloon is $r \sin \beta \operatorname{cosec} \left[\frac{\alpha}{2} \right]$

[CBSE 2023]



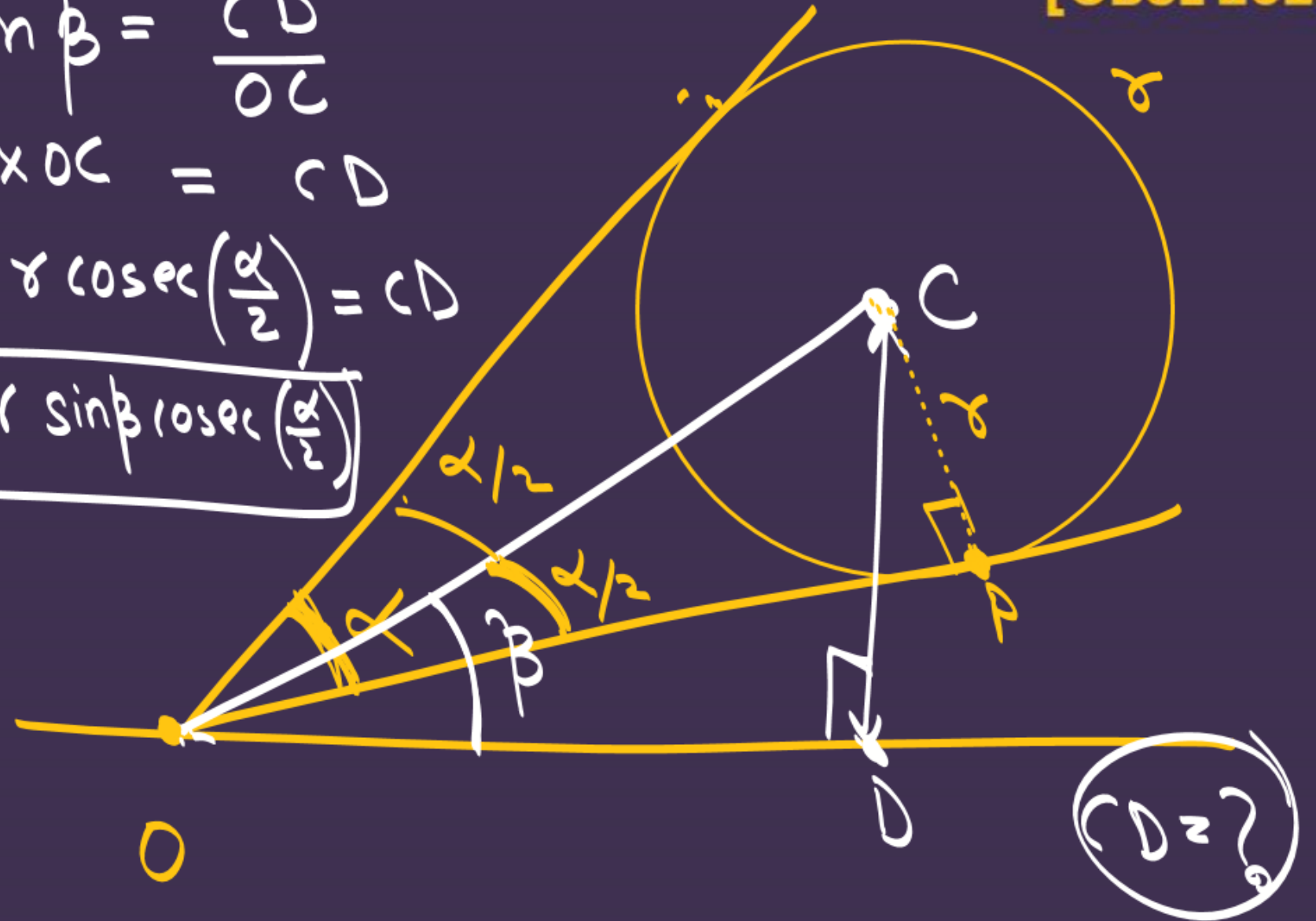
$$\sin\left(\frac{\alpha}{2}\right) = \frac{r}{OC}$$

$$OC = \frac{r}{\sin\left(\frac{\alpha}{2}\right)}$$

$$OC = r \operatorname{cosec} \left(\frac{\alpha}{2} \right)$$

$$\sin \beta = \frac{CD}{OC}$$
$$\sin \beta \times OC = CD$$
$$\sin \beta \times r \operatorname{cosec} \left(\frac{\alpha}{2} \right) = CD$$

$$CD = r \sin \beta \operatorname{cosec} \left(\frac{\alpha}{2} \right)$$



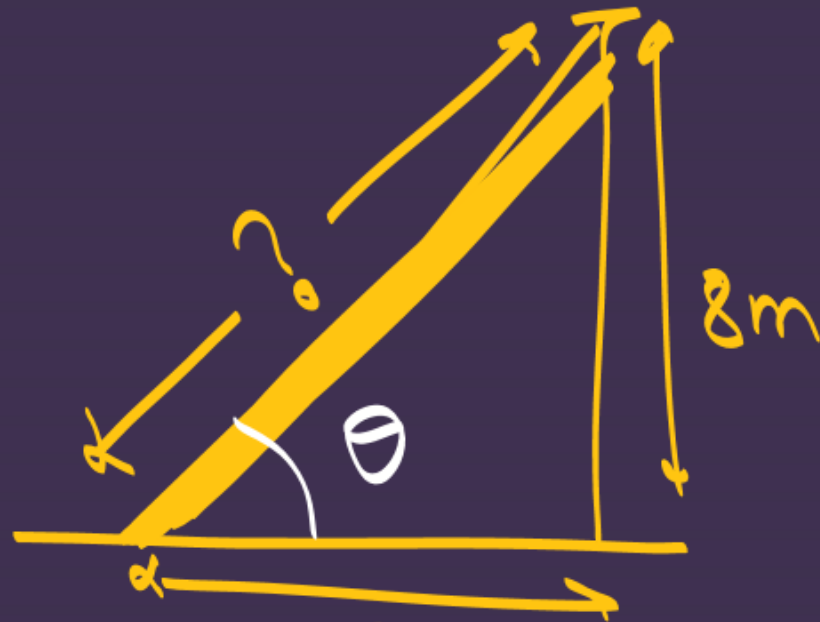
#LP: Directions: In the following questions, a statement of assertion (A) is followed by a statement of reason (R). Mark the correct choice as:

- (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).**
- (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).**
- (c) Assertion (A) is true but reason (R) is false.**
- (d) Assertion (A) is false but reason (R) is true.**

#LP: Assertion (A): A ladder leaning against a wall, stands at a horizontal distance of 6 m from the wall. If the height of the wall up to which the ladder reaches is 8 m, then the length of the ladder is 10 m. (T)

Reason (R): The ladder makes an angle of 60° with the ground. (F)

[CBSE 2025]



Pythg

$$H^2 = P^2 + B^2$$
$$H^2 = 8^2 + 6^2 \Rightarrow H^2 = 100 \Rightarrow H = 10\text{m}$$

$$\tan \theta = \frac{8}{6}$$

$$\tan \theta = \frac{4}{3}$$

$$\tan 60 = \sqrt{3} \quad (F)$$

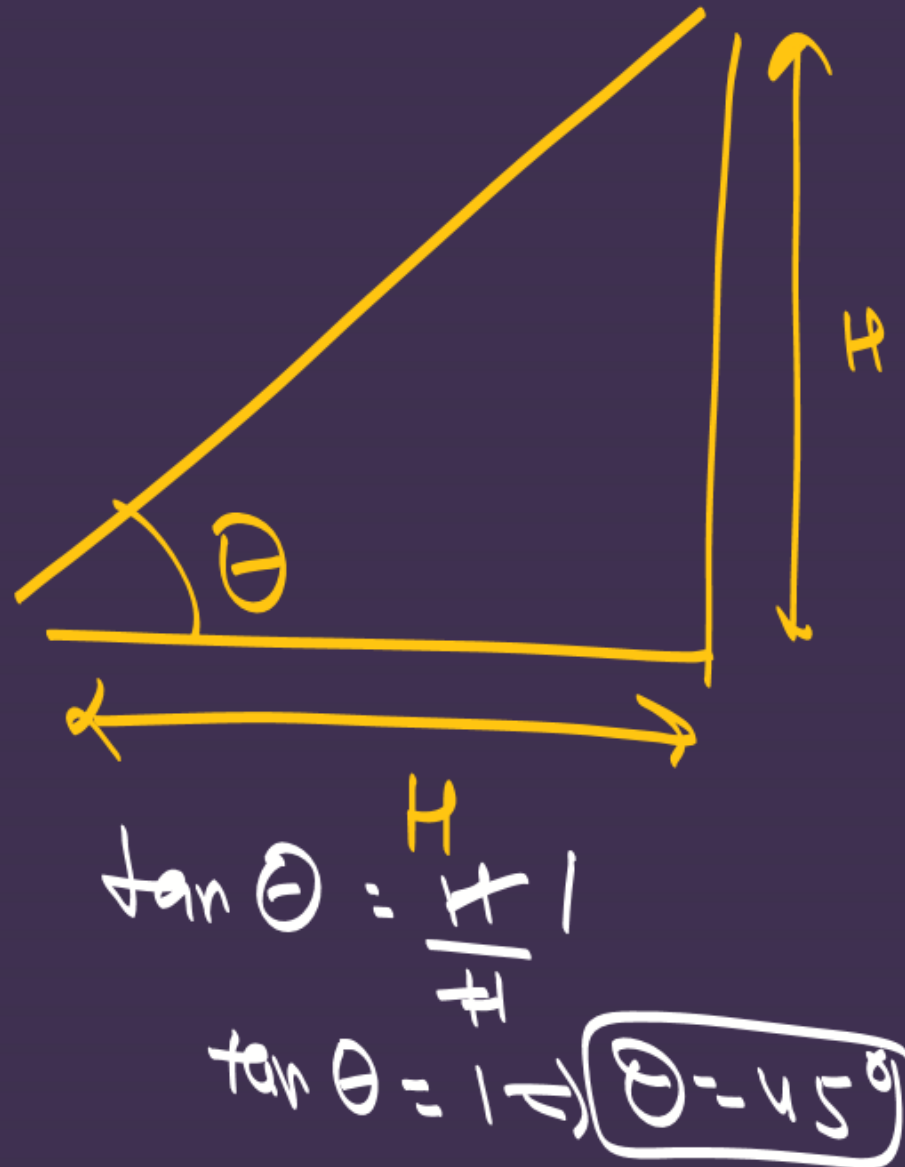
(C) ✓

#LP:

Assertion (A): If the length of shadow of a vertical pole is equal to its height, then the angle of elevation of the sun is 45° . τ

Reason (R): According to Pythagoras theorem, $h^2 = l^2 + b^2$, where h = hypotenuse, l = length and b = base. τ

[CBSE 2024]

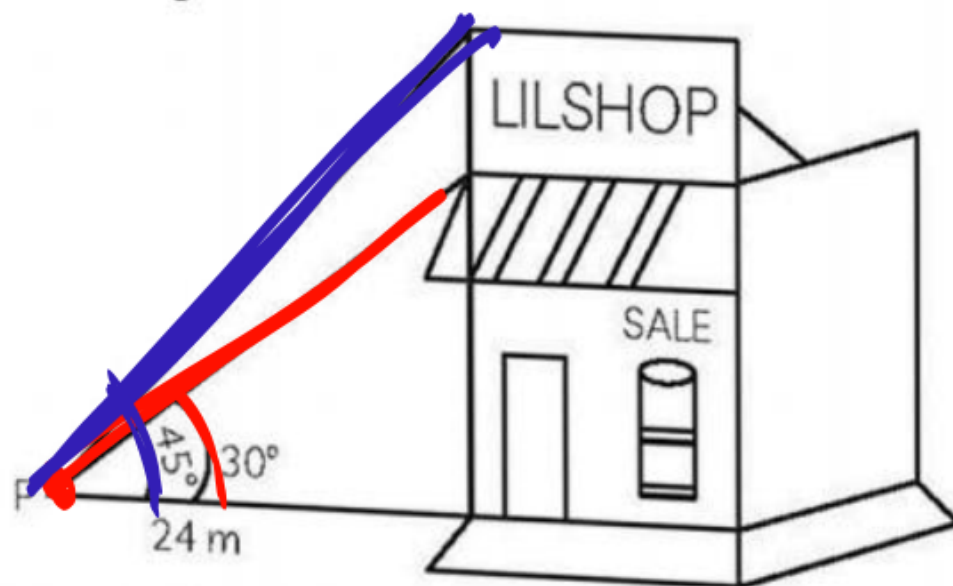


#LP:

Anita purchased a new building for her business. Being in the prime location, she decided to make some more money by putting up an advertisement sign for a rental ad income on the roof of the building.



From a point P on the ground level, the angle of elevation of the roof of the building is 30° and the angle of elevation of the top of the sign board is 45° . The point P is at a distance of 24 m from the base of the building.



(i) Find the height of the building (without the sign board). $\rightarrow AB$ (2)

OR

Find the height of the building (with the sign board) $\rightarrow AC$ (2)

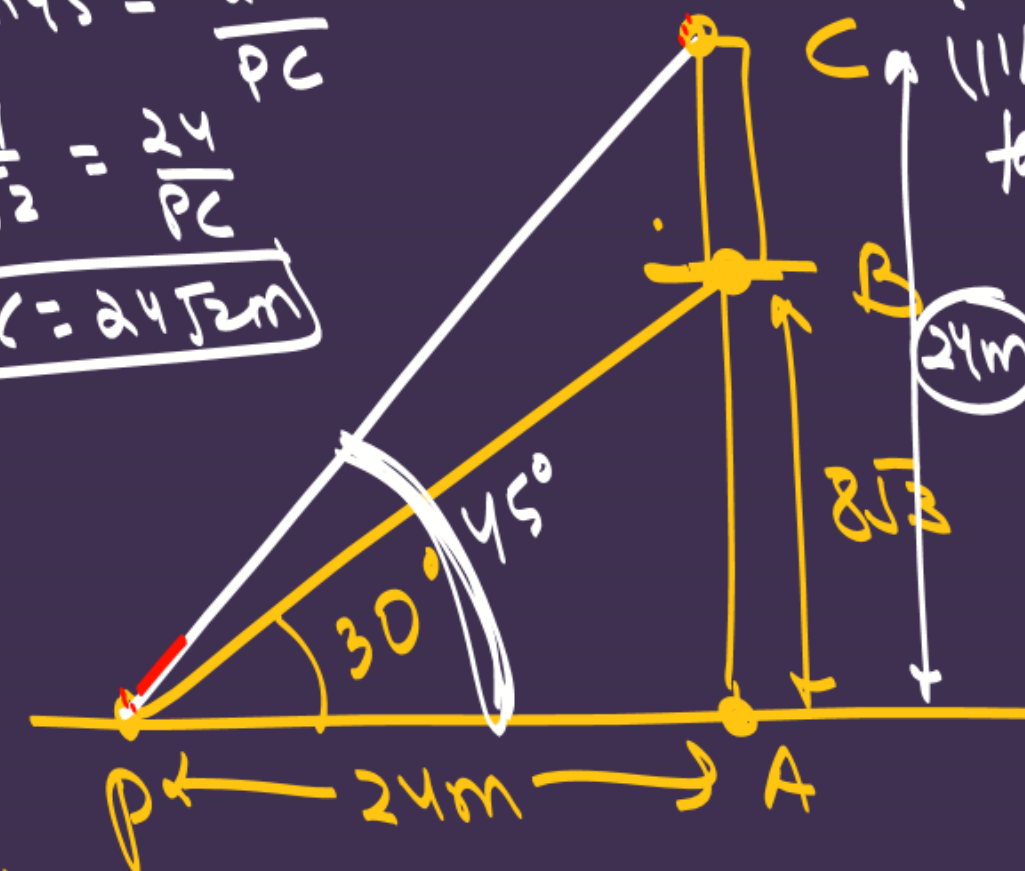
(ii) Find the height of the sign board. $CB = 64 - 8\sqrt{3}$ (1)

(iii) Find the distance of the point P from the top of the sign board. PC (1)

$$\text{iii) } \sin 45 = \frac{24}{PC}$$

$$\frac{1}{\sqrt{2}} = \frac{24}{PC}$$

$$PC = 24\sqrt{2} \text{ m}$$



$$\text{iii) } \tan 45 = \frac{AC}{24}$$

$$1 = \frac{AC}{24}$$

$$AC = 24 \text{ m}$$

$$\text{i) } \tan 30 = \frac{AB}{24}$$

$$\frac{1}{\sqrt{3}} = \frac{AB}{24} \Rightarrow AB = \frac{24}{\sqrt{3}} \rightarrow 8\sqrt{3}$$

3/14

Completed!

Thank you Goodieess !!

